UNIVERSITY

UNIVERSITY COMMON COURSE

Course Title: BUN 100: Greening for Sustainability

Course Purpose

The course seeks to introduce students to the university vision, mission and its niche areas of 'green economy for sustainability". It aims at imparting knowledge and skills that empower students to think and act green in critical areas of individual life and societal living for sustainable development.

Course Description

Definition of concepts- greening, sustainability; Linear Economy- strengths and weaknesses; Circular (Green) Economy; Policy and Legal framework on the green concept; Greening in Agriculture, Environment - renewable energy, biodiversity conservation, Built Environment, Waste Management, recycling; Climate Change and the Green Economy; Science and technology – green technology, green infrastructure; Business and entrepreneurship – green jobs, resource use efficiency; Arts and Culture – preservation of indigenous heritage, indigenous industries, indigenous foods and cuisines, indigenous fashion and indigenous sports, games and music; green behavior - Emotional Intelligence, Interpersonal Skills, Time management, Peaceful coexistence; Gender Mainstreaming and Affirmative Action for sustainability, Health and Wellbeing – nutrition, fitness and exercise, non-communicable diseases, Positive Mental Health, Drugs and substance Abuse, Sexual Reproductive Health; Education, Training, Research and Innovation – integrating digital technology in training and teaching. Digital Economy; Integrating digital economy and green economy. Regional integration, Current Trends and opportunities.

Course Delivery

The course will be delivered through a Multidisciplinary approach using:

Face-to-face Learning;

Blended learning;

Case-Based Learning (CBL);

Problem Based Learning (PBL)

Course Evaluation/Assessment

The course is mandatory, but the marks will not contribute towards graduation. Modes of assessment may include but not limited to:

- Multiple Choice and Short-Answer Questions preferably using on-line Examination platform,
- Essay writing
- Short Project Report.

Teaching and Learning Recourses

Core Texts

1. Adrian C. Newton, *An Introduction to the Green Economy*, Routledge, 2014.
2. Carmen Nadia, "Integrating Digital Economy and Green Economy: Opportunities for Sustainable Development", in *Theoretical and Empirical Researches in Urban Management*, vol. 6, No. 1 (Feb. 2011), pp. 33-43, available at : https://www.jstor.org/stable/10.2307/24873273

Recommended Texts

1. Godwell Nhamo and Vuyo Mjimba (eds.), *Sustainability, Climate Change and The Green Economy*, Africa Institute of South Africa, 2016.

Staff: To be sourced from all schools (i.e. SASS, SPAS, SBE, SOE) and sections.

BUN 100: GREENING FOR SUSTAINABILITY

SPAS

INTRODUCTION AND DEFINITION OF CONCEPTS

Greening: it is a situation where a person or organization becoming more aware of environmental issues such as using smaller cars and dimly lit rooms, eating greens etc

Sustainability : the processes and actions through which humankind avoids the depletion of natural resources, to keep an ecological balance that does not allow the quality of life of modern societies to decrease.

Biodiversity: Biodiversity is the variability among living organisms from all sources, including terrestrial, marine, and other aquatic ecosystems and the ecological complexes of which they are part; this includes diversity within species, between species, and of ecosystems. Biodiversity forms the foundation of the vast array of ecosystem services that critically contribute to human well-being. Biodiversity is important in human-managed as well as natural ecosystems.

Green economy

- According to the United Nations Environment Programme (UNEP) **A green economy** is one that results in improved human well-being and social equity, while significantly reducing environmental risks and ecological scarcities. A green economy can be thought of as one which is low carbon, resource efficient and socially inclusive.
- In this type of economy, growth in income and employment are driven by both public and private investments that reduce carbon emissions, enhance resource efficiency, and prevent the loss of biodiversity and ecosystem services.
- key component of this economy is that economic development views natural capital as a key economic asset and as a source of public benefit
- The overall aim of a transition towards a green economy is to enable economic growth and investment while increasing environmental quality and social inclusiveness.
- The green economy approach is a shift away from the short-term understanding of environmental considerations as a cost factor that constrains economic growth and reduces competitiveness. it views these considerations as fundamental to the long-term sustainability of economic growth.
- Overall, the green economy is one that results in improved human well-being and social equity, while significantly reducing environmental risks and ecological scarcities

Green Growth

- The benefits of economic growth were gained at the expense of the environment. The green paradigm indicates that the previous approach to growth, such as polluting and degrading the environment first and then cleaning up and restoring the environment afterwards, must be suspended.
- Instead, a new path should advocate sustainable development that protects the environment. Green growth has emerged as a new development paradigm to respond to the traditional unsustainable

energy and carbon-intensive models that are based on economic growth without consideration for the environment.

- Therefore, **Green growth** means fostering economic growth and development, while ensuring that natural assets continue to provide the resources and environmental services on which our well-being relies.
- To do this, it must catalyse investment and innovation which will underpin sustained growth and give rise to new economic opportunities.
- Green growth is relevant to developed countries which need to retrofit their resource-consuming industries and lifestyles and to developing countries that can avoid copying damaging development pathways.
- They can 'leapfrog' old solutions and adopt new technologies and ideas. Meanwhile, in developed countries, the challenge of transitioning towards the green economy will be to change lifestyles and reduce consumption of natural resources to sustainable levels.
- In developing countries, the challenge will be to stimulate economic growth, so the green economy coincides with sustainable development.

Characteristics of green growth

- More effective use of natural resources in economic growth
- Valuing ecosystems
- Inter-generational economic policies
- Increased use of renewable sources of energy
- Protection of vital assets from climate-related disasters
- Reduced waste of resources

Objectives of the Green Economy

Green economy is characterized by three objectives:

- **Improving resource-use efficiency**: a green economy is one that is efficient in its use of energy, water, and other material inputs.
- **Ensuring ecosystem resilience**: it also protects the natural environment, its ecosystems, and ecosystem flows.
- **Enhancing social equity**: it promotes human well-being and a fair burden sharing across societies

Improving Resource-Use Efficiency

- The rising global population will increase demand for water, energy, food, and natural resources, in turn increasing consumption of material resource globally
- Therefore, there is the need to sustain economic growth in the longer term while keeping negative environmental impacts under control and preserving natural resources
- By improving resource efficiency, it means economies can use the Earth's limited resources in a sustainable way to create more with less while reducing residual waste to close to zero.
- A key aspect of improving resource-use efficiency is decoupling economic activity from consumption and environmental impacts.
- There are **two dimensions to decoupling**:
    - **Resource decoupling** means reducing the rate of use of resources per unit of economic activity and impact **<u>decoupling means</u>** maintaining economic output while reducing the negative environmental impact of any economic activities undertaken.
    - **Combined, relative decoupling** of resources or impacts means the growth rate of the resources used or environmental impacts is lower than the economic growth rate, so that

resource productivity is rising. Absolute reductions of resource use happen when the growth rate of resource productivity exceeds the growth rate of the economy.

Ensuring Ecosystem Resilience

- An ecosystem is a dynamic complex of plant, animal, and microorganism communities and the non-living environment interacting as a functional unit. An ecosystem can be the distribution of organisms, soil types, drainage basins, or even depth in a body of water.
- Ecosystem services are the benefits humans obtain from ecosystems. These include provisioning services such as water and food; regulating services such as flood and disease control; cultural services such as spiritual, recreational, and cultural benefits; and supporting services such as nutrient cycling that maintain the conditions for life on Earth.
- **Ecosystem resilience** meanwhile refers to the capacity of an ecosystem to recover from disturbance or withstand ongoing pressures. It is the measure of how well an ecosystem can tolerate disturbance without collapsing into a different state. An ecosystem's ability to absorb or recover from impacts and its rate of recovery depend on the biology and ecology of its component species or habitats; the conditions of these individual components; the nature, severity, and duration of the impacts; and the degree to which potential impacts have been removed or reduced.

Enhancing Social Equity

- The integration of the social aspect, or human well-being, in the green economy is fundamental given the importance of basic resources—food, water, energy, and materials—as well as ecosystem services for human subsistence needs.
- Regarding green jobs and inclusiveness, job opportunities in an inclusive green economy will be significant but they need to be identified early and the education and skills training needed to fill them will need to be a priority for any government engaging in a transition towards this new economy.
- Overall, the extent of the creation of green jobs will be seen as an indicator of competitiveness: the higher an economy's creation of green jobs, the better it is posed to compete in a future where economies are designed to produce wealth and income without creating environmental risks and resource scarcities.
- On the issue of equity and natural resources, the green economy depends on the capacity of disadvantaged groups to organise collectively; engage in contestation, advocacy, and bargaining; and be part of broader coalitions of change.
- To facilitate active citizenship, the green economy's governance arrangements need to be sensitive to the issues of diversity, representation, and space for negotiation and ensure policies are not dominated by narrow or elite interests.
- Governments can also cultivate an enabling environment for participation and empowerment through education and training as well as institutionalizing accountability mechanisms and basic rights and freedoms of association, expression, information, and redress

GREENING IN AGRICULTURE & BIODIVERSITY CONSERVATION

Smart farming

The term **smart agriculture** refers to the usage of technologies like Internet, sensors, location systems, robots and artificial intelligence on your farm. The goal is increasing the quality and quantity of the crops while optimizing the human labor used. Example technologies used in smart agriculture are:

- **Precision irrigation and precise plant nutrition**
- **Climate management and control** in greenhouses

- **Sensors** – for the soil, water, light, moisture, for temperature management
- **Software platforms**
- **Location systems** – GPS, satellite, etc
- **Communication systems** – based on mobile connection
- **Robots**
- **Analytics and optimization platforms**

The connection between all these technologies is the Internet of Things – this is a mechanism for connectivity between sensors and machines, resulting in a complex system that manages your farm based on data received. Thanks to this system, farmers can monitor the processes on their farms and take strategic decisions remotely – from their tablet, phone or other mobile device – without being on the open fields, in their greenhouse, orchard, vineyard, etc.

Types of Smart Farming Technology

Smart farming technologies (SFTs) are divided into three main categories

1) **Data acquisition technologies**: this category contains all surveying, mapping, navigation and sensing technologies.
2) **Data analysis and evaluation technologies**: these technologies range from simple computer-based decision models to complex farm management and information systems including many different variables.
3) **Precision application technologies**: this category contains all application technologies, focusing on variable-rate application and guidance technologies.

Data Acquisition Technologies

The SFTs for recording and mapping field and crop characteristics are divided into the categories below:

- Global navigation satellite systems technologies (in fact these technologies record the actual position which can be used for different purposes such as guidance, mapping etc.)
- <u>Mapping technologies</u>
    - **Elevation Maps**
        - Elevation is a critical layer in precision agriculture (PA) because it is very useful to help farmers understand yield response. It influences soil formation, water movement and cropping aspects.
        - It can determine waterlogged areas, erosion risk, drainage restrictions, and often is related to soil type.
        - Elevation maps can help to identify how topography can affect agronomic results in a field and of course to level the field.
        - Using this information, it is possible to produce (i) contours and topography maps, (ii) 3-D modelling of ponding risk, runoff and velocity maps, (iii) farm layout designs, (iv) contour bank design, drainage plans and on-ground implementation and (v) cut and fill land levelling designs.
    - **Soil Mapping**
        - Soil sampling is vital to collect information about soil texture (sand, silt, clay contents), availability of nutrients for crops to grow (N, P, K, Ca, Mg, pH, lime) and other soil chemical properties (organic matter, salinity, nitrate, sulphate, heavy metals).
        - it can be used to identify soil compaction, moisture content and other mechanical and physical soil properties.
    - **Yield Mapping**
        - Yield mapping or yield monitoring is a technique in agriculture of using Global navigation satellite systems (GNSS) data to analyse variables such as crop yield and moisture content in a given field.

- Data acquisition of environmental properties (Camera based imaging, NDVI measurements, soil moisture sensors)
  - **Camera Based Imaging**
    - **RGB Cameras**-Red, Green and Blue (RGB) cameras combine the colours red, green and blue to depict the range of colours that exist in the environment and in the agricultural fields.
    - Through the use of digital image analysis, a significant correlation of the red, green and blue of digital images with the protein content of soybean plants
    - **LiDAR Sensors**-LiDAR sensors (Light Detection and Ranging) are instruments that measure the distance from the target by laser. This technology has been used to study the phenotypic variation by creating three-dimensional models of plants.
    - **ToF (IR) Cameras**-Time of Flight (ToF) cameras have the ability to produce shaped and incoherent infrared light in the space. Smart sensors at pixels of the camera record the reflected light and calculate the time to return. These cameras are used to measure the distance between the corn plants in a row.
    - **Light Curtains**-Light curtains are a new system that is used to study the phenotypic traits. The system consists of a couple of bars which are placed in parallel. light curtains are used to measure the height and leaf area of corn, tomato, barley and oilseed rape plants.
    - **Thermal Cameras**-Thermal cameras have the ability to generate images related to the ambient temperature. Thermal cameras have been used to study the phenotypic variance for predicting water stress of plants, to detect diseases and pathogens and for the ripening of fruits.
- Machines and their properties Global navigation satellite systems (GNSS) technologies

Smart farming processes

1. **Data collection**
   The sensors installed at all critical places in the farm gather and transmit data about the soil, air, etc
2. **Diagnostics**
   The data collected is analyzed by the system and conclusions are made regarding the status of the object or process monitored. Potential problems get identified.
3. **Decision making**
   Based on the problems identified in the previous steps, the software platform and/or a human managing the platform decides on actions that need to be taken.
4. **Actions**

   - The actions identified in the previous step are performed. A new measurement on the soil, air, moisture, etc is performed by the sensors and the whole cycle starts again.
   - The result from this automated smart farming process is – **high precision and 24/7 control**, eventually leading to **considerable savings in all key resources used** – water, energy, fertilizers, time spent by strategic people, time spent by lower-qualification human resources.
   - Smart farming can **save up to 85% in water consumed and up to 50% in energy consumed**. They also report **up to 40% increase in crop yield**, while reducing the cost of fertilization and chemical treatment, and up to **60% less losses resulting from human error**.

Smart Farming Technologies

The SFTs can be classified according to the following parameters:

- Farming system type, which is divided into three major farming systems:
  - **Organic farming:** The farming system that relies on biological control and mechanical cultivation to maintain soil productivity and pest control.

- **Extensive farming**: Low energy input farming that uses small inputs of labour, fertilizer and capital relative to the land being farmed and where conventional practices are carried out.
    - **Integrated farming:** The farming system that use good agricultural practices and or integrated crop management strategies to produce high quality and certified agricultural products.
- Cropping systems type, which can be divided into the main clusters that correspond to main cropping environment within EU and have distinctive differences among themselves:
    - Arable crops
    - Forage crops
    - Orchards
    - Vineyards
    - Field vegetables

Value addition on agricultural produce

- The wide range of topography, soil quality and climatic conditions that describes Kenya makes her agriculture advantageous for producing a large number of crop and non-crop products. The possibility of processing some of these products to value added items signifies sizeable potential for the development of the agricultural sector in Kenya.
- Though Kenya has a strong raw material base, it has been unable to tap the real potential for processing. The processing units based on grains, horticultural products, livestock products and fish have ample opportunities.

Agro Processing and Value Addition

- **Agro processing** is the conversion of agricultural product to substances which have particular textural, sensory and nutritional properties using commercially feasible methods. It is necessarily a process of value adding activity to agricultural production and thus makes agriculture a more effective contributor of industrial growth.
- **Value addition** involves transformation of the raw materials into final consumer goods or intermediate goods and thus results in increase in value. Or it's a process of changing or transforming a product from its original state to a more valuable state.
- The value adding processes range from simple preservation to production of high value products. For example, a farmer cultivates paddy on his farm and the paddy plants produce paddy, straw, husk, bran, and rice kernel. Paddy has a potential of supporting a number of processing industries such as rice mills, solvent extraction plant for rice bran oil, processing of husk for variety of products and straw paper or board mills and the processing of these raw materials opens up large value addition possibilities.
- Agro processing adds value to the agricultural, horticultural, livestock and fisheries products by using various value addition techniques which enhances their shelf life. It leads to diversification of agricultural activities, improves value addition opportunities and creates surplus for export of processed agro products.
- Agro processing often goes simultaneously with agricultural diversification as perishable products like meat, milk, poultry, fruits and vegetables, often termed as High Value Products (HVPs), are in greater demand of processing. Faced with the current economic realities, farmers worldwide are searching for new options of surviving, as well as expanding their agro business.

Creating values

Occurs with actual or perceived value to a customer for a superior product or service

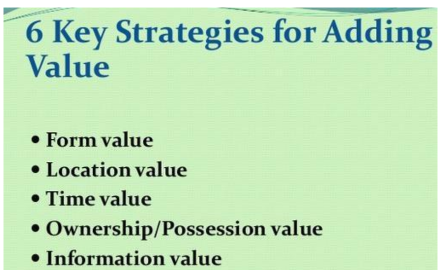
- Innovative new products

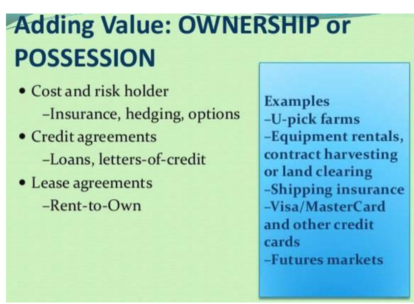
- Enhance a product's characteristics
- Enhance service
- Create brand names
- Develop unique customer experiences

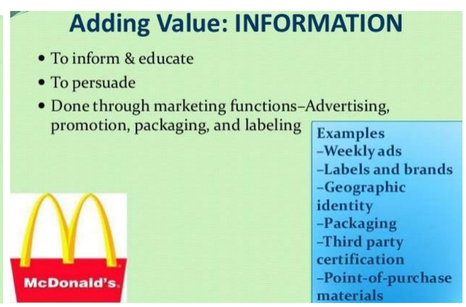
Capturing value

Changing the distribution of value in the food/fiber production chain is meant to 'capture' more of the consumer through:

- **Direct marketing**-selling products to the consumers, selling homemade soaps and lotions to the general public.
- **Vertical integration-**one producer or business owns the product from beginning to end. This producer doesn't sell the product until the consumer purchases it.
- **Producer alliances**- individual/companies from the same level of the food chain consolidate in order to produce and market a superior product
- Cooperative efforts- individuals or companies pool their products in order to increase bargaining power.







BUN 100: Greening for Sustainability.

KEY CONCEPTS

- Greening - A **green economy** is an economy that aims at reducing environmental risks and ecological scarcities, and that aims for sustainable development without degrading the environment. The process of achieving such an economy is 'Greening'
- Sustainability – Refers to avoidance of the depletion of natural resources in order to maintain an ecological balance. It has **three** pillars: the economy, society, and the environment. Sustainability is essentially a balance of the three pillars to ensure that the Earth's biosphere and human civilization co-exist.
- A **sustainable economy** is one that is resilient and provides a good quality of life for everybody.

KEY CONCEPTS

- Linear Economy (LE): An **economy** based on 'take-use/make-dispose‘. The approach of **LE** involves raw material transformation into finished goods and waste products. The used goods are disposed or accumulated as waste.
- Circular Economy (CE) – An economy that aims at eliminating waste and ensures the continual use of resources; it is regenerative by design. The **CE** Mantra: reduce, reuse, recycle, recover, repair.

A Circular Economy is an Economy with No Waste!

The Need for Greening

The previous revolutions: Agrarian and Industrial

- Before 1760: Agrarian economy.
  - Powered by human and animal labour
  - Individual & community life centered around farming
  - **Biomass** is the principle source of energy.
- An industrial revolution occurs when new technologies and novel ways of perceiving the world trigger a profound change in economic and social structures (Klaus Schwab)
- Three distinct industrial revolutions have occurred to date and we are on the threshold of a 4th one.

The 1st Industrial revolution

- The age of mechanical production (1760 – 1890)
- Powered by **steam** (steam engine & machine tools)
- Transportation: Steam ships and railroads
- The factory emerges as the new center of community life
- Unskilled labour (plenty and cheap – included child labour) and skilled labour (Middle class)
- Urbanization and Industrialization progress rapidly – Urban (cities with factories) and rural settings

The 2nd Industrial revolution

- The age of science and **mass production** (1890 – 1950)**.**
- Powered by **fossil fuels** (esp. petroleum) & **electricity**
- Scientific (Lab) principles were brought into the factory
  - The Assembly line. Powered mass production e.g. Ford cars
- Specialization of labour
- Transportation: Motor vehicles, airplanes and electric rail
- Communication – radio and telephones
- Chemical fertilizers (synthesis of ammonia)
- Electric lighting
- Rural – urban migration (USA: 40% urbanites in 1900, 6% in 1800)

The 3rd Industrial revolution

- The age of digital revolution (starting from 1950s)
- Semiconductors, mainframe & personal computing, the Internet
- Shift from analog to digital technologies
- Automated production (Electronics & IT) and global supply chains
- Disrupted assembly line labour - High unemployment rates
- Great advances in communication
- Scientific breakthroughs: robotics, artificial intelligence, miniaturized sensors, 3D printing et.c.
- Emphasis began on renewable energy sources

4th Industrial revolution

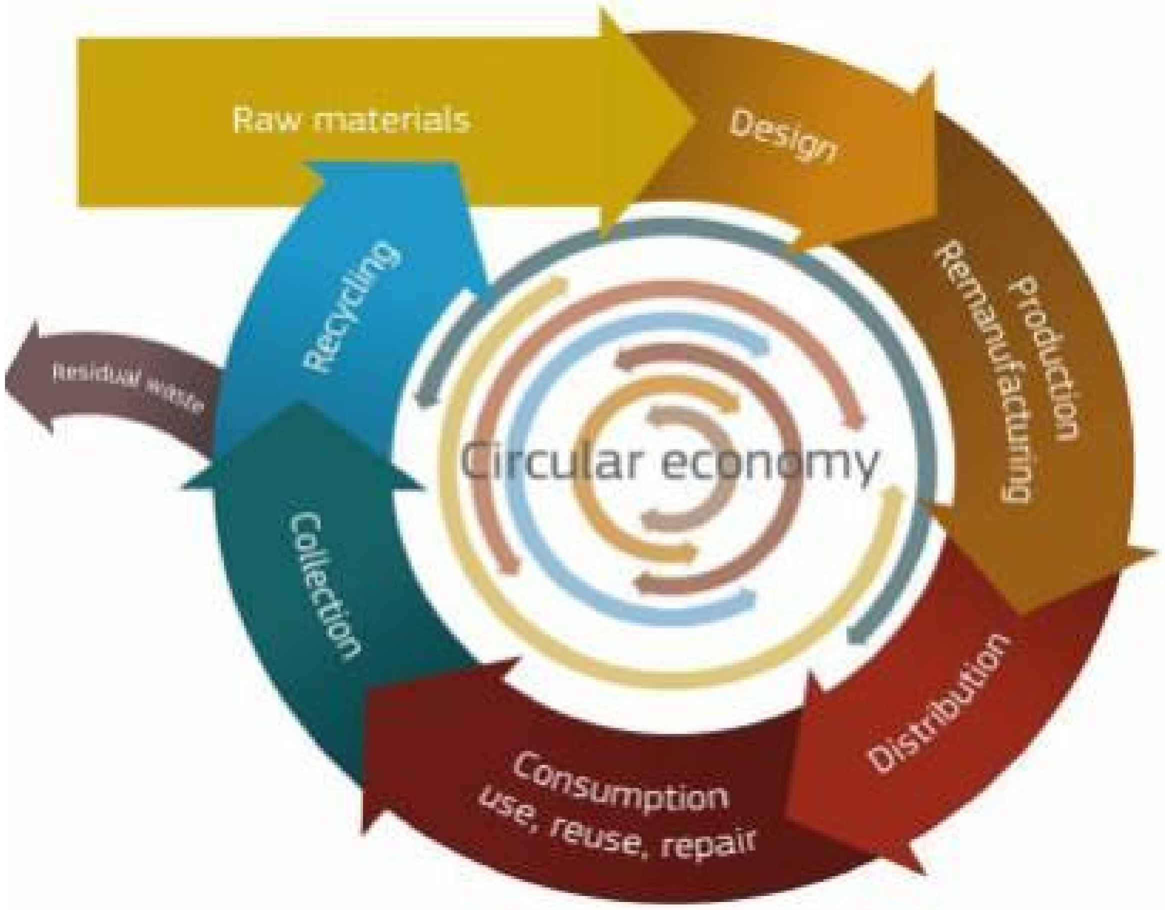
- The age of **Cyber-physical systems** –(just beginning for some nations. Others are yet to experience it)
- Includes greater & novel embedment of technology into humans/societies e.g. genome editing, novel forms of machine intelligence, novel materials et.c.
- It could be described as an ‘assembly line’ of digital technologies
- It builds on the advances of the 3rd industrial revolution
- The shape and direction it will take is yet to unravel
- All the past revolutions were premised on the **Linear Economy**

Linear vs Circular Economy

Linear economy - waste

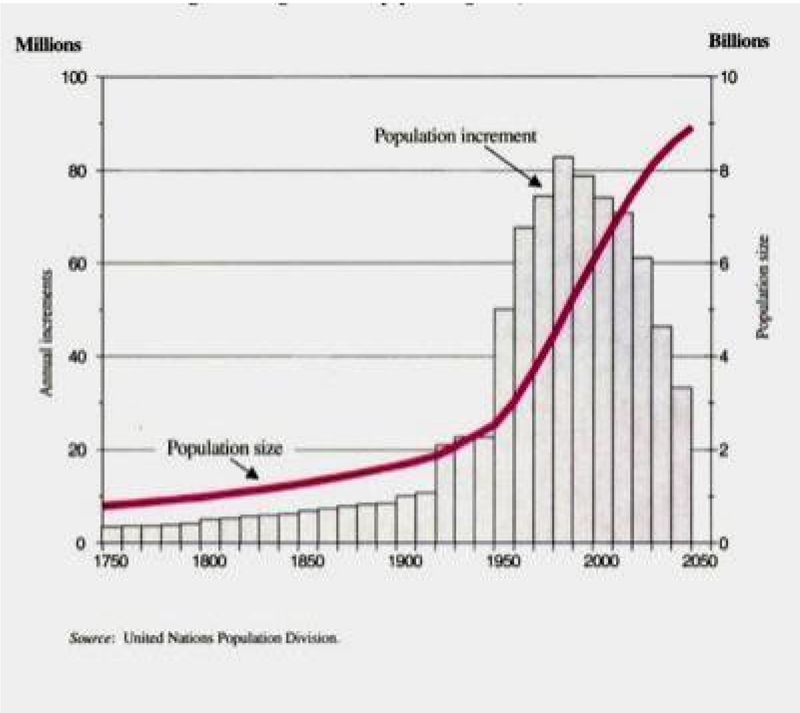
- Take → Use → Dispose
- Waste results in pollution: Water, soil, air, Food
- Diminishing non-renewable natural resources – fossil fuels, minerals et.c.
- Human waste – High unemployment levels

Circular economy – No waste



Population Growth !

1960- 1975   a billion people added
1975-1987 another billion added

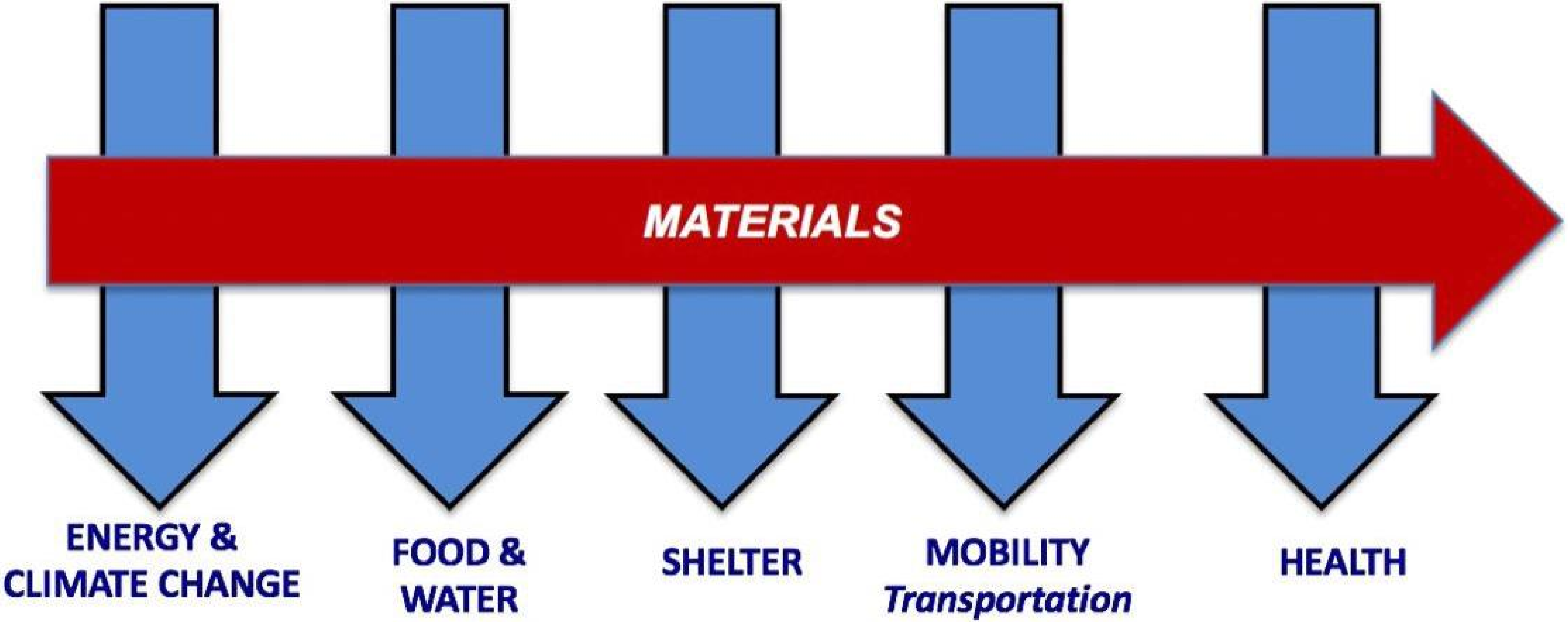


Entered the 20th Century with 1.6 billion

Exited the 20th Century with 6.1 billion

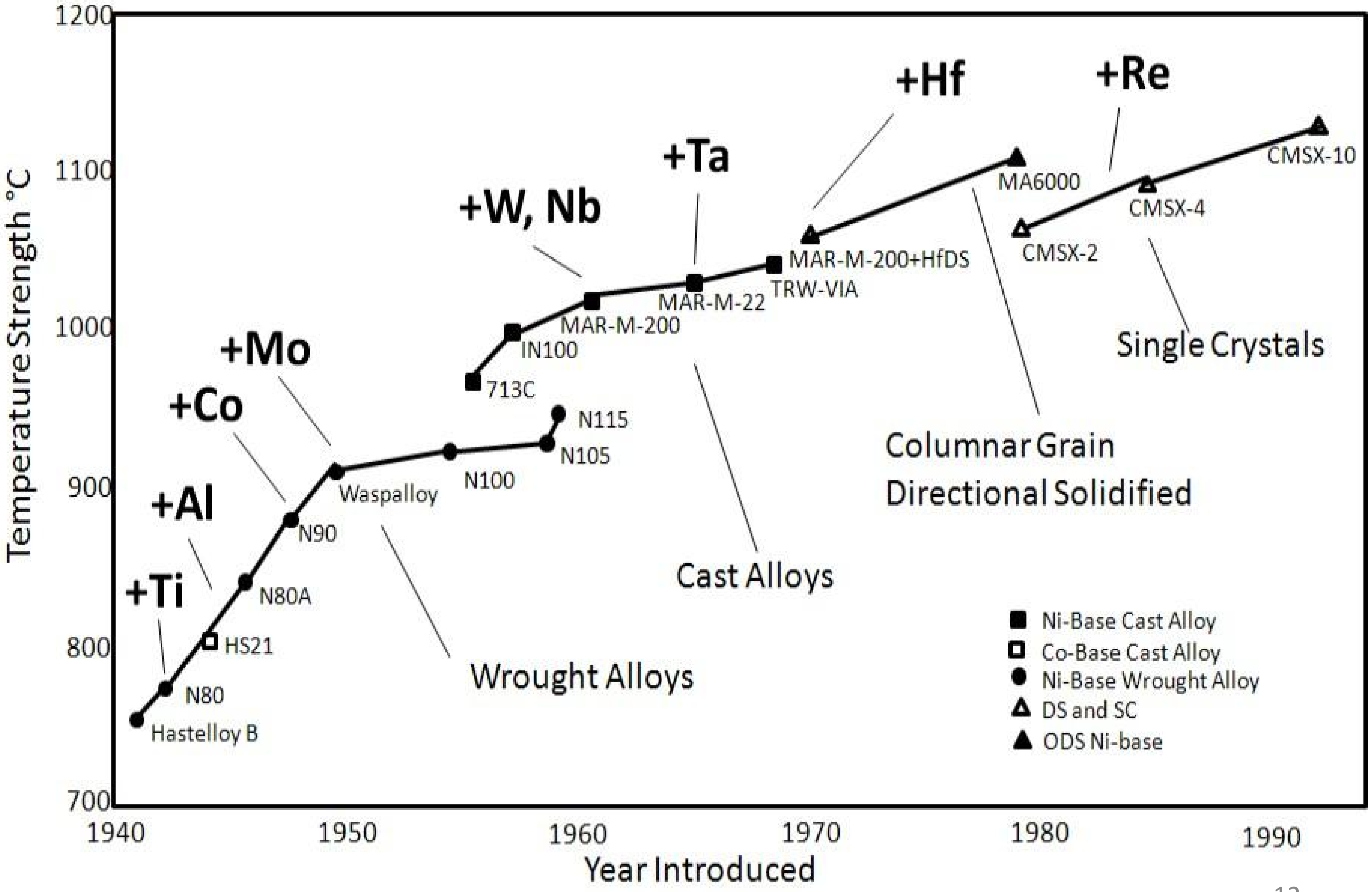
SUSTAINABLE DEVELOPMENT

SOCIETAL NEEDS - CHALLENGES

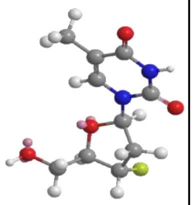


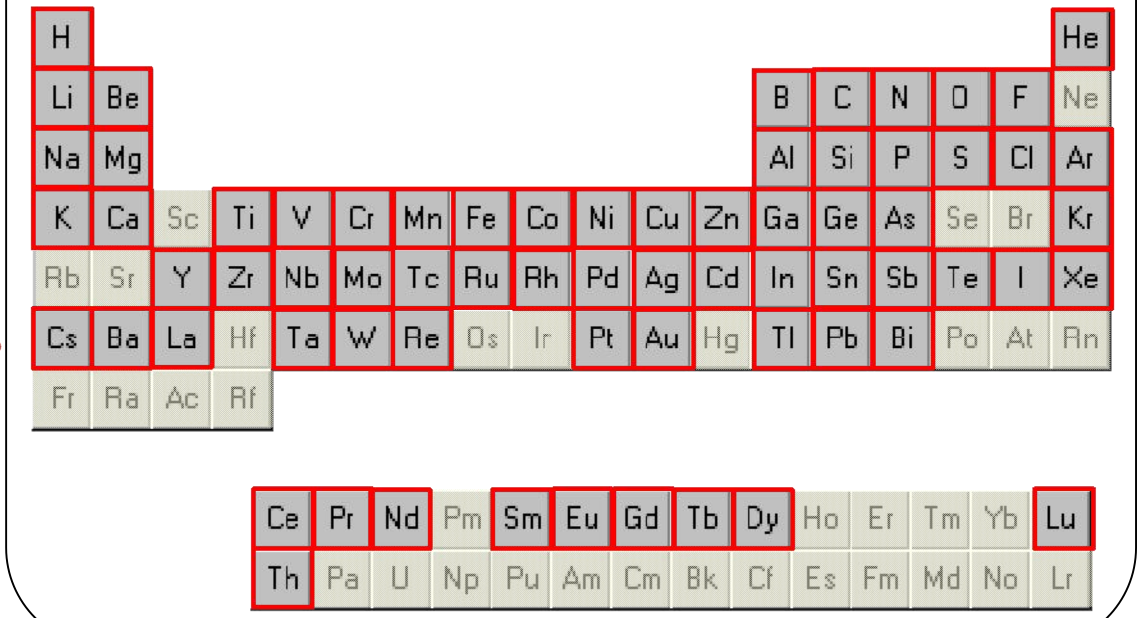
DEMANDS OF INDUSTRIALIZATIONON THE PERIODIC TABLE

1. The GE 90 Jet Engine super alloy composition over time

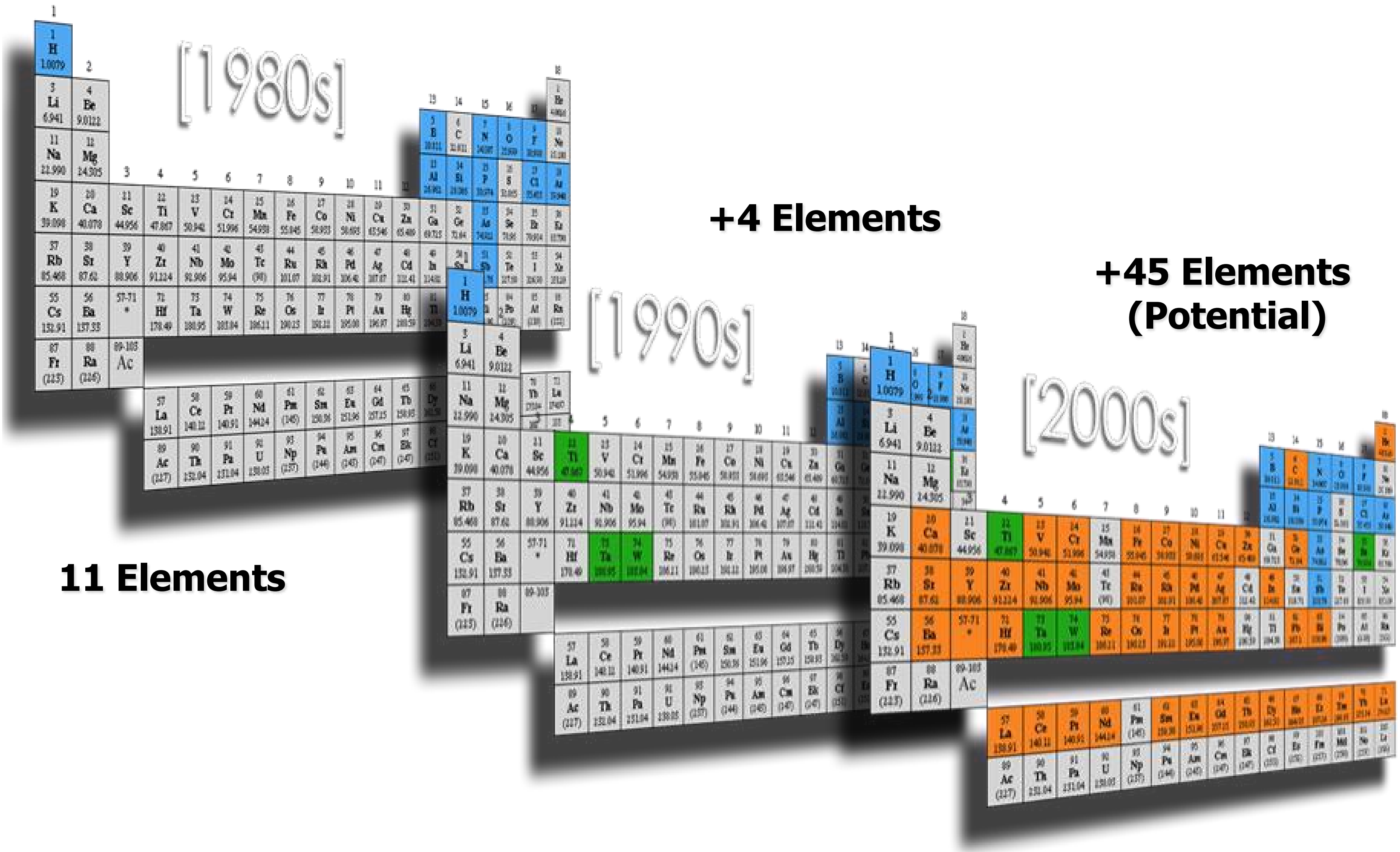


2. Elements Used in General Electric Healthcare Products



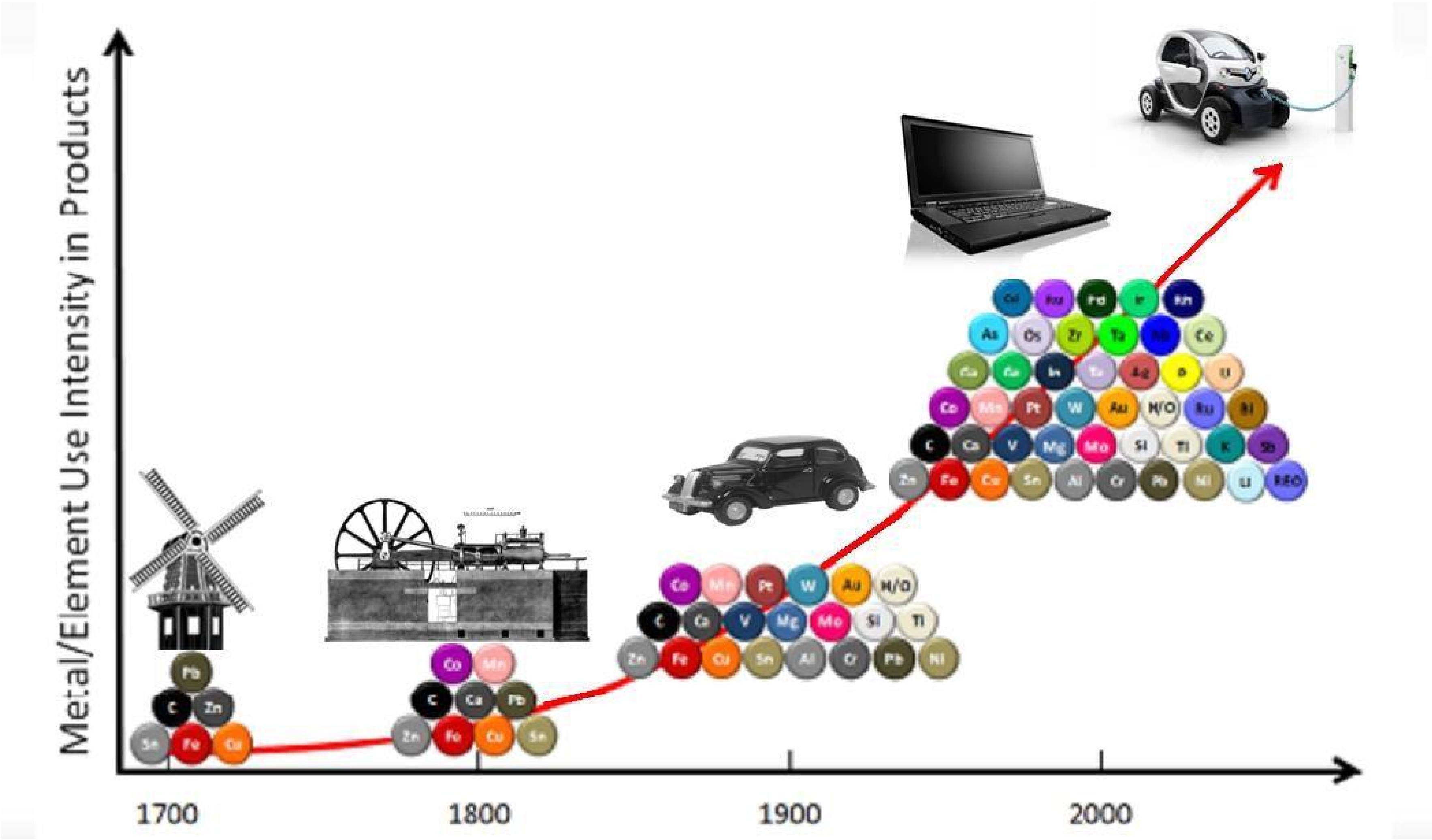


3. Computer Chip Elemental Contents



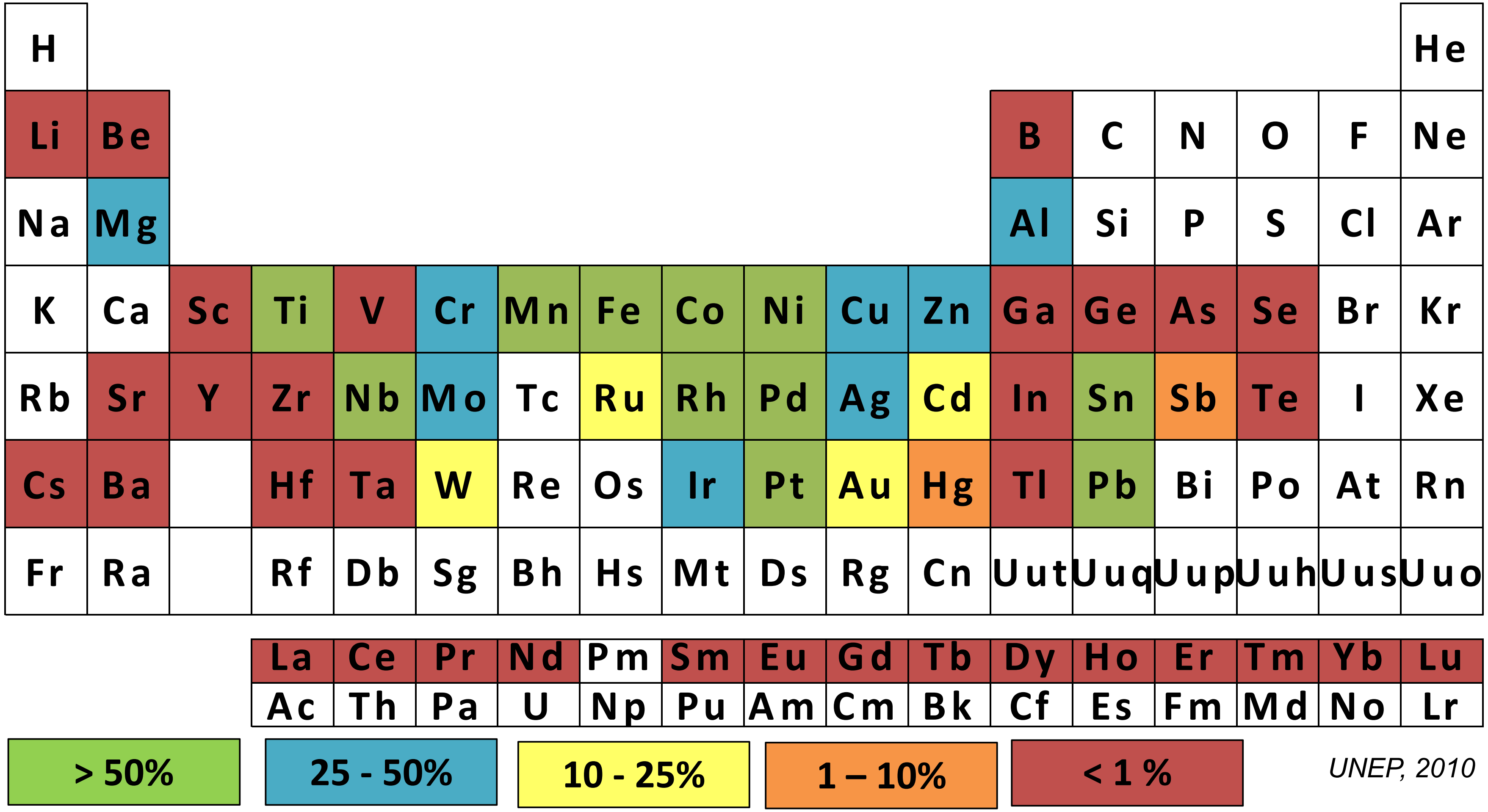
Source: T. McManus, Intel Corp., 2006

Complexity in elements we use is increasing



Based on: Achzet et al., Materials critical to the energy industry, Augsburg, 2011

End-of-life Recycling Rates



UNEP, 2010

The challenge

- Most of these elements are in danger of getting exhausted – some within the next 30 – 50 years
- Very few phoshate deposits are left in the world – only in Northern Africa – and may not last the next 30 years. This has serious consequences for agriculture.
  - Agricultural nutrients: C, N, P.
  - C and N are renewable (from air) but P is not.
  - Depletion of phosphate rock will mean no further source of phosphate fertilizers
- Most of the elements occur in nature as a family and are also used in many products as a family leading to huge losses in waste!

A Lesson from Nature we SHOULD mimic

Natural Ecosystem		Industrial System
EQUILIBRIUM	**INDICATOR OF WELL-BEING**	**GROWTH**
Uses a Few Elements (mostly C, N, O, H)		**Uses most of the Periodic Table**
Is Cyclic Materials circulate and transform continuously		**Is Linear** Transforms materials into products & waste
Subsystems Have evolved & use waste as a resource		**Lack of Subsystems** ∴ Can't use waste as a resource
Closed Loop *No waste.* Each subsystem provides sustenance for others		**Open Loop** *Waste.* Destructive of sources which it depends

WHAT WE NEED IS A CRADLE TO CRADLE APPROACH

TAKE AWAY MESSAGE

SAUSAGE

- **REDUCE PRODUCTION WASTE**
- **REDUCE POST CONSUMER WASTE**
- **DESIGN AND MANUFACTURE for DIASSEMBLY**
- **CREATE VALUE FROM WASTE …**

Strategy: RECOVER AND REUSE

The Circular Economy and Future of investments

- A Circular Economy is a World in Which There is No Waste
- No Material Waste - Requires strategic thinking about Minerals, Waste Recovery and Recycling
- No Energy Waste - Requires thinking on our Energy Resources – both active and passive
- No Human Waste - Requires us to think about the People of the Earth and their potential to lead fulfilling lives
- Advent and utilization of cyber-physical systems

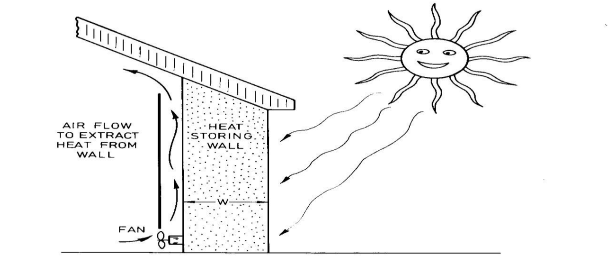
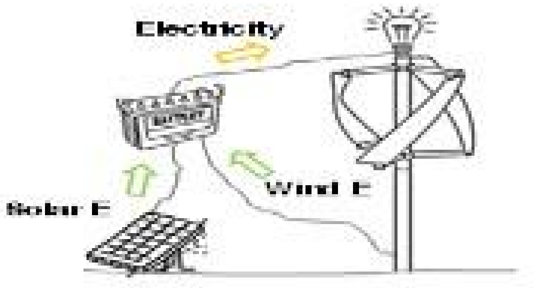


FIG. 6.32 A heat-storing wall. The sun shines on the outside during the day; heat is extracted from the inside at night. The heat diffusion time through the wall must be about 12 hours.



WHAT IS THE MOST PRECIOUS AND VALUABLE ASSET WE HAVE IN AFRICA?

HUMAN CAPITAL

What do we do?

- **GOLDEN OPPORTUNITY – Embrace the circular economy and ride with the fourth industrial revolution**
- **INVEST IN HUMAN CAPITAL *(The most amazing renewable energy source that will fuel the Continent)***
- Go Green on a LARGE SCALE
- CREATE VALUE

SOLID WASTE MANAGEMENT

Sources of waste

Solid waste can be categorized by its origin. Common waste generation categories include residential, commercial, institutional, and industrial enterprises.

1. **Residential.** Includes all types of households, such as single-family homes, apartments, and other types of formal and informal housing. Waste generated by this sector usually includes food and organic waste; textiles paper and cardboard; and small portions of glass, rubber, leather, and metals. A small portion of plastic is also included; this fraction tends to increase with economic growth and globalization. Household hazardous waste is a subset of residential waste that includes chemicals such as paints, solvents, cleaning agents, batteries, and electronics.

2. **Commercial.** Includes office buildings, shopping malls, hotels, airports, restaurants, and markets. Markets, restaurants, canteens, and hotels tend to have waste streams with a high percentage of food waste and other organic components. Offices, hotels, and warehouses tend to generate a large quantity of recyclables such as paper, cardboard, plastic, and glass.

3. **Institutional.** Includes schools, medical facilities, and prisons. Institutional facilities often generate large quantities of paper. Some institutions – including hospitals and schools – also generate high volumes of food waste. Medical facilities generate hazardous waste, which should not be handled with general solid waste.

4. **Industrial.** Includes manufacturing or industrial process facilities. Packaging components, lunchroom and restroom wastes, textiles, scrap metal, wood/lumber scrap, masonry/concrete, and similar wastes are typical waste products from industrial facilities. The type of waste produced is related to the type of industry, but is typically produced in high quantities. Industries usually produce both hazardous and non-hazardous waste, so it is a best practice to ensure that the hazardous waste is managed based on the country's legal requirements and is not mixed in and collected with non-hazardous solid waste.

Why is waste management important?

Inadequate solid waste management can impact cities and their residents in many ways. These impacts can generally be categorized into three categories:

- **Human health-**The improper handling of waste can impact human health (e.g., decomposing organic waste attracts rodents, insects, and stray animals). In some cities, human faecal matter and urine are not separated from solid waste, which attract insects and germs that spread disease (e.g., typhoid, cholera).
  Mosquitos also pose a concern when they breed in solid waste (e.g., used tires); mosquitos can be vectors for diseases such as malaria, dengue, and the Zika virus.
  Mismanaged solid waste and open dumpsites can lead to environmental contamination of surface and groundwater, which are common sources of drinking water.
  Uncontrolled burning of waste may result in emissions of air pollutants including dioxins, furans, black carbon, heavy metals, and particulate matter, many of which can be toxic for human health. For populations living in direct contact with or close proximity to waste disposal sites, these health effects can be particularly severe.

- **Environmental -** Inadequate control of leachate, water that filters through waste and draws out chemicals, at disposal sites can lead to environmental contamination of soils and waterbodies, impacting local ecosystems.

Mismanaged waste is also a threat to stray animals and wildlife as animals may try to consume waste that contains food residue or scraps. and global climate.

Waste disposal sites release methane, which contributes to the formation of ground-level ozone. In addition, methane is a greenhouse gas that contributes to climate change.

- **Socioeconomic -** Inadequate solid waste management can be costly, both in terms of direct expenses and indirect costs.

  Mismanaged solid waste systems are a missed opportunity for economic growth, including increased property values and tourism benefits from having clean streets and beaches.

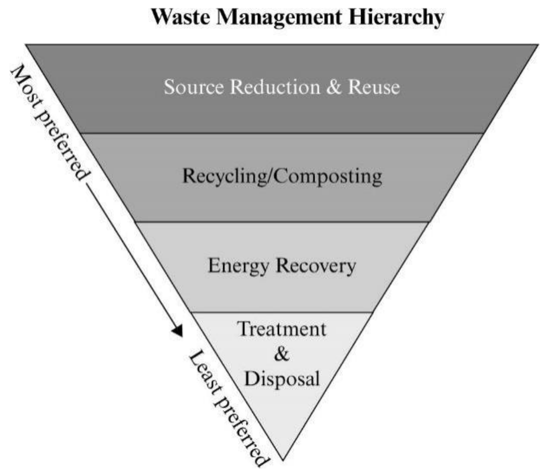
  Improved solid waste management can especially benefit highly vulnerable populations through cost savings on public health systems by preventing respiratory issues, skin diseases, and other health care concerns associated with inadequate solid waste management.

Common Challenges in waste management

1. Limited financial resources and capacity.
2. Limited access to and technical knowledge of equipment.
3. Limited technical expertise and awareness of best practices.
4. Limited staff capacity
5. Political turnover. Changes in administrations can result in projects being shut down or radically altered by incoming officials
6. Lack of planning and evaluation at both national and county levels
7. Limited or lack of vertical and horizontal government coordination
8. Limited available land for waste management facilities

Approaches to waste management

- Effective waste separation and collection programs are a critical component of an integrated solid waste management system. These activities involve arrange of stakeholders, from individual households to collection fleet operators.
- Many cities have found it important to establish clear methods of communications and coordination among those groups. Effective waste separation, collection, and transportation also involve a variety of types of infrastructure, including receptacles for separating and storing waste before it is collected; and vehicles such as carts, bicycles or tricycles, and trucks.
- Uncollected waste can lead to serious human and environmental impacts including spread of diseases, local flooding due to clogged drainages, marine litter that can affect marine organisms, local water and air pollution.
- No single solid waste management approach is suitable for managing all waste materials and in all circumstances. Local governments should work to create a plan that meets its specific needs and conditions.
- The United States Environmental Protection Agency developed a solid waste management hierarchy (see below) in recognition of this reality. This hierarchy provides a general ranking system for the various solid waste management strategies from most to least environmentally preferable; and places emphasis on reducing, reusing, and recycling.



Solid waste management hierarchy

Source Reduction and Reuse

- Source reduction, also known as waste prevention, refers to reducing the amount of waste generated. Reducing waste at the source is the most environmentally preferred strategy.
- Individuals can reduce the amount of waste they generate by purchasing long-lasting and reusable products, or seeking out products that have been designed with waste reduction in mind.
- Prevention and minimization of waste, and the processes and practices intended to reduce the amount of waste produced, is a best practice for solid waste management systems. Reducing waste and reusing materials are not only environmentally beneficial, but can deliver public health benefits and save money.
- Because waste prevention avoids waste generation, it is the most cost-effective and preferred solid waste management activity. Preventing or minimizing waste conserves resources (e.g., by reducing collection and transportation costs), protects the environment, and prevents the release of greenhouse gases.
- The following strategies can be applied in the prevention and minimization waste production;
  1. Redistributing food that would otherwise be wasted.
  2. Packaging material in the waste stream can be minimized by seeking products with minimal packaging, and instituting fees for plastic and paper bags, or banning them.
  3. Disposable product use could be minimized by encouraging the purchase of durable, long lasting goods.

<u>Recycling and Organic Waste Management</u>

- Recycling refers to collecting and processing materials that would otherwise be disposed of as waste and turning them into new products. Such materials include; paper, glass, aluminium, steel, plastics, used motor oil, batteries, tires, and electronic waste.
- Organic waste in the solid waste stream is generally divided into two categories:
  - ***Food loss and waste* -** Food waste includes unused produce from pre-consumption sources (e.g., markets and restaurants) and food left over after consumption. Food loss includes unused products from the agricultural sector (e.g., unharvested crops).
  - ***Green waste* -** Green waste includes waste from gardens, landscaping, and tree trimming.
- Organic waste management deals with the diversion and treatment of organic waste through composting and anaerobic digestion (AD).
  - **Composting -** Composting is the controlled decomposition of organic materials in the presence of oxygen. Composting requires three general steps:
    1. Combining organic waste types, such as wasted food, yard trimmings, and manure;
    2. Adding wood chips, shredded paper, or other bulking agents to accelerate the breakdown of organic waste; and

3. Allowing the compost to stabilize and mature through a curing process.

- **Anaerobic digestion** - Involves the breakdown of organic materials by microorganisms in the absence of air. The products of the AD process include biogas, an energy source that contains mostly methane and carbon dioxide, and digestate. Digestate is the material that is leftover after organic materials are anaerobically digested. Digestate is rich in nutrients and can be used as fertilizer for crops.

Energy Recovery

- Energy recovery is the conversion of non-recyclable materials into useable heat, electricity, or fuel through a variety of processes. This process is often called waste-to-energy.
- Converting non-recyclable materials into electricity and heat generates an energy source and reduces carbon emissions by offsetting the need for energy from fossil sources, and reduces methane generation from landfills.
- Waste-to-energy plants have a high upfront investment cost and are costly to operate and maintain.
- Additionally, toxic emissions from such units have to be controlled. When coupled with effective end-of-pipe air pollution controls (i.e., controls placed on a facility that treats gases before they enter the environment) and waste disposal techniques, these plants can potentially reduce both waste volumes and greenhouse gas emissions. However, adequate financing plans and effective pollution controls are key factors to consider before planning a waste-to-energy facility as a viable solid waste management option.

Treatment and Disposal

- Prior to disposal, treatment can help reduce the volume and toxicity of waste.
- Treatments can be;
  - physical (e.g., shredding)
  - chemical (e.g., incineration) and
  - biological (e.g., Anaerobic digestion).
- Landfills are an important component of an integrated solid waste management system. Waste that cannot be prevented or recycled should be disposed of in properly designed, constructed, and managed landfills, where it is safely contained to limit its environmental impacts.
- Methane gas, a by-product of decomposing waste, can be collected and used as fuel to generate energy. After a landfill is capped, the land may be used for other purposes such as recreational sites.

BIODIVERSITY CONSERVATION

- ***Biodiversity*** refers to the various life forms that exist on earth, including animals, plants, microorganisms, and the entire ecosystem they live in.
- Biodiversity is fundamental as it ensures natural sustainability of all life on earth not only for the current population but also for the future generations.
- However, biodiversity continues to be threatened and in consequence, it affects the survival of humans.
- ***Biodiversity conservation*** is all about protecting all organisms and species within their natural habitats with the aim of ensuring intragenerational and intergenerational equity.
- Activities such as habitat fragmentation, human disturbance, and habitat loss have to be adequately curtailed to enrich biodiversity conservation measures.
- Below we discuss different solutions to the various threats to biodiversity.

Various Threats to Biodiversity and Their Solutions

1. Habitat loss and deforestation

- The dramatic alteration of habitats directly threatens biodiversity. When such habitats are lost due to deforestation and other anthropogenic activities such as mining, the respective environments are unable to provide shelter, food, water, or breeding grounds for the living organisms.
- In other words, it leads to unhealthy and unbalanced ecosystems that result in the loss of biodiversity and extinction. Deforestation, in particular, is associated with the destruction of about 18 million acres of forest habitats annually, damaging the ecosystems on which countless species depend on for survival.

Solution

- Curbing the problem of deforestation and habitat loss mostly depend on policy and implementation of laws.
- Corporations and companies can take up the wise practice of refusing to use paper and timber products that encourage deforestation.
- Likewise, awareness creation should be done so that consumers can desist from supporting companies or manufacturers that use timber and paper to make their products and most importantly, those that utilize unsustainable natural resource extraction or manufacturing processes.
- Governments and regulatory agencies should also take the lead in ensuring the enactment of stronger forest protection laws and policies.
- Individuals and organizations can also participate by support environmental conservation through charities and creating awareness.

2. Climate change

- The global climatic changes throughout the history of the planet have definitely modified life and ecosystems in the planet. As an outcome, crucial habitats have been destroyed and a number of species have gone extinct with a huge majority at the verge of extinction.
- It therefore means that if the global temperatures continue to change drastically, especially due to anthropogenic activities that accelerate the process, the threats to biodiversity will continue to expand as ecosystems and species will not be able to adapt.
- For instance, the decreasing Arctic sea ice and the increasing ocean temperatures are to blame for the changes in vegetation zones and deterioration of marine wildlife. Besides, the ecosystems and species that cannot cope die out.

Solution

- Climate change is mostly exacerbated because of human activities with regards to destroying carbon sinks and their dependency on using fossil fuels. Hence, if effective actions can be taken to

reduce the amount of carbon footprint, the world can be assured of a better tomorrow and less worries about climate change.
- Individuals, organizations and industries need to reduce their carbon footprints and they should equally participate in awareness creation.
- Cities and international governments can also charge for carbon emissions and enact policies that curtail activities which destroy the carbon sinks.

3. *Overexploitation of resources*

- On the account of the ever-rising human population, there has been a correlational increase in demand for manufactured products, essential goods and services.
- The high demands of these things have resulted in overfishing, overhunting, over-harvesting and excessive mineral resource extraction which has highly contributed to biodiversity loss.
- Mineral extraction, poaching, excessive logging and other forms of resource exploitation for profit has heightened the risks of species extinction. It has also altered natural habits therefore destroying food chains and interfering with the ecological balance.

Solution

- Continued awareness creation and conservation are the main strategies for managing overexploitation, particularly pertaining to overfishing, over-harvesting and poaching.
- Relevant environmental protection agencies and governments also need to implement rules curtailing practices that cause overexploitation of resources.
- Individual effort should aim towards being mindful of the products we consume and buy.

4. *Nutrient loading*

- As the agricultural sector continue to expand and serve towards attaining the world's food security, it has also more than doubled dependence on the use of fertilizers on a profitable scale.
- Accordingly, the use of fertilizers beyond limits has contributed to increased level of nitrogen and phosphorous nutrients in the natural ecosystems. As much as the nutrients exists naturally in all ecosystems, the manufacturing of artificial fertilizer with reactive nitrogen and phosphorus nutrients to increase crop productivity has altered the ecological balance over time thereby threatening the survival of ecosystems.
- Particularly, the survival of species that flourish in phosphorous or nitrogen-poor environments are increasingly threatened.
- Furthermore, leaches and entry into water systems have resulted in increased eutrophication and the creation of anoxic (oxygen deficient) zones in marine habitats.

Solution

- To address the problem of nutrient loading, considerable improvements are needed to ensure the effectiveness of the nitrogen and phosphorus fertilizers used within production systems. The artificial production of these fertilizers should not only aim at meeting the global food demands but also decreasing environmental problems.
- The GreenFacts Foundation suggests that if the globe's cereal production systems can attain a 20% gain in nitrogen-use efficiency, it would cut back about 6% of the global production of reactive nitrogen.
- Farmers should also reduce their dependency on using high quantities of nutrient fertilizer and seek alternatives such as leaving the land fallow after harvesting to encourage nutrient recycling.

5. *Environmental pollution*

- Pollution has continued to harm the biosphere by releasing and depositing toxic chemicals into the atmospheric, terrestrial and marine systems. With the high levels of pollution every year, it is gradually disrupting the Earth's ecosystems as the chemicals released potentially influences species' habits and ecosystems.

- Pollution has also depleted ozone levels, created dead zones in marine habitats due to toxicity and acid rains, altered species feeding and breeding habits, and even caused the death of many species due to oil spills or the consumption of plastic and other toxic substances.

Solution

- There are a number of ways for curbing pollution. Anti-pollution laws and policies at the local, state and international level are the most practical for curbing pollution as they play a critical role in restricting pollution.
- Individuals can also take a number of initiatives to fight water, air and land pollution by embracing effective actions such as conserving energy at home, recycling, use of safe and non-toxic products, and using public transport.
- Awareness creation and advocacy is as well essential. It can be done through the media, online educative forums, and in various institutions to make people realize and understand the causes and consequences of environmental pollution.
- The use of renewable and green energy sources can also lower dependence on fossil fuels thereby reducing greenhouse gas emissions that lead to global warming and the formation of acid rain.

6. Invasive species

- Invasive species are the non-native species that invade normal and healthy ecosystems and threaten the survival of the native species either by attacking them or competing for the habitat's resources. Accordingly, they upset the native biota and ecosystems thereby causing extinctions and massive threats to biodiversity.
- According to the GreenFacts Foundation, nearly 40% of all animal extinctions since the 17th century are associated with invasive alien species. Similar reports also indicate that 80% of the threatened species in the Fynbos biome of South Africa are endangered as a result of invading alien species. Besides, the report emphasizes that the cumulative environmental biodiversity losses of more than $100 billion in the UK, US, South Africa, Brazil, India, and Australia are because of invasive pests.

Solution

- The problem of invasive alien species is global and for this reason, it requires intervention at the local, state and international level. Local authorities and states need to establish systems to manage and prevent invasive alien species through risk assessments as a strategy of predicting the possibility of species becoming invasive. The assessments should also aim at determining the potential ecological damages and put in place effective preventive measures to counter the likely environmental impacts.
- Therefore, creating systems to stop the introduction of invasive alien species even before it happens, quickly eliminating newly detected invaders, and effectively monitoring new invasions are the most efficient strategies. International bodies and scientist can assist in research and information quantification by using more creative means such as Google street view and other advanced technological techniques

……

FITNESS AND EXERCISE

Definitions:
Fitness is a state of health and well-being and, more specifically, the ability to perform aspects of sports, occupations and daily activities. Physical fitness is generally achieved through proper nutrition, moderate-vigorous physical exercise and sufficient rest
Exercise is a physical activity that is planned, structured, and repetitive for conditioning the body. The exercise consists of cardiovascular conditioning, strength and resistance training, and flexibility.

Purpose of fitness and exercise
Exercise is essential for improving overall health, maintaining fitness, and helping to prevent the development of obesity, hypertension, and cardiovascular disease. Surveys conducted by the Centers for Disease Control and Prevention (CDC) indicate that 61.5 per cent of youths aged 18 to 35 years do not participate in any organized physical activity (for example, sports, dance classes) and 22.6 per cent are not physically active during their free time. Approximately 30 per cent of youths and adults aged 19 – 40 years are overweight and 15 per cent are obese.
A sedentary lifestyle and excess caloric consumption are the primary causes of this increase in overweight and obesity; regular exercise is considered an important factor in controlling weight. Overweight and obese children and adolescents are at higher risk of developing several medical conditions, including the following:

- Asthma
- Diabetes
- Hypertension
- Orthopaedic complications, such as hip and knee pain and limited range of motion
- Cardiovascular disease
- High cholesterol
- Sleep apnea
- Psychosocial disorders, such as depression, negative body image, and eating disorders

Clinical studies have shown that regular exercise has numerous benefits, including the following:

- Preventing weight gain and maintaining a healthy weight
- Reducing blood pressure and cholesterol
- Improving coordination
- Improving self-esteem and self-confidence
- Decreasing the risk of developing diabetes, cardiovascular disease, and certain types of cancer
- Increased life expectancy

Exercise and fitness consist of cardiovascular conditioning, strength and resistance training, and flexibility to improve and maintain the fitness of the body's heart, lungs, and muscles.

Types of exercises and fitness
There are four different types of exercises and fitness that can be incorporated into your exercise routine:

A. Cardiovascular/Aerobic
B. Anaerobic
C. Joint flexibility

D. Muscle strength and endurance

Cardiovascular or Aerobic exercise and fitness

Cardiovascular conditioning involves moderate to vigorous physical activity that results in an elevated heart rate for a sustained period. Cardiovascular exercise and fitness target the muscles' ability to make the best use of oxygen so that they can produce energy for movement.

Regular cardiovascular exercise improves the efficiency of the functioning of the heart, lungs, and circulatory system. For adults, aerobic exercise within a target heart rate range calculated based on a maximum heart rate by age is recommended.

For healthy persons within the university gap (19 – 35 yrs), cardiovascular exercise that elevates the heart rate to no greater than a maximum heart rate of 200 beats per minute is recommended. In general, the American Heart Association recommends at least 90 minutes of moderate to vigorous physical activity every day for teenagers and adults. Cardiovascular exercise and fitness are often considered to be the most important type of physical exercise and fitness due to the wealth of health benefits it offers. These are directly related to the condition of the lungs and heart and has been known to significantly increase both the length and quality of life.

A strong heart and a healthy set of lungs is a requirement for clear blood vessels that supply muscles with oxygen. It helps to maintain favourable body composition, as well as improved stamina.

Anaerobic exercise and fitness

Anaerobic exercise and fitness are that which is directly related to short, powerful bursts of energy such as that required for sprinting, powerlifting and short, fast movements. Exercises related to anaerobic fitness should be intense enough to cause the formation of lactate, eventually enhancing strength, speed and power for non-endurance activities.

There are five different types of anaerobic activity:

- Weight-lifting
- Sprinting
- Plyometrics (movements associated with increased muscular power)
- Isometrics (activity done from a static position)
- High-intensity interval training requires repetitive short and powerful movements.

Joint flexibility

Flexibility is important to improve and maintain joint range of motion and reduce the likelihood of muscle strains. In addition to this, it accounts for the lengthening of muscles across joints to facilitate a bending motion. Joint flexibility will vary from person to person, but can still be cultivated with the right workout routine to increase their range of movement.

Flexibility is especially important for university students and staffs by engaged in vigorous exercise (running, competitive sports). Stretching is best performed following a warm-up and/or after an exercise session or sport. One activity that promotes flexibility that is increasing in popularity is yoga, in the form of yoga classes or exercise videos.

Flexibility results in several benefits including fewer injuries when exercising, suppression of joint pain, enhanced posture and improved balance as well as better strength.

Muscular strength and endurance

Muscular strength is a type of fitness directly related to muscle density and endurance, allowing athletes to perform more repetitions while lifting heavier weights. Endurance and strength training has several positive impacts on the body, including increased bone density to reduce the risk of osteoporosis. Strength and resistance training increases muscle strength and mass, bone strength, and the body's

metabolism, generally, It increases a person's ability to work against resistance by maximising the force that can be applied against a load.
Strength training can be performed with or without special equipment. Strength/resistance training equipment includes handheld dumbbells, resistance machines (Nautilus, Cybex), and elastic bands. Strength training can also be performed without equipment; exercises without equipment include pushups, abdominal crunches, and squats.

Precautions to take during exercise
Before a person begins any exercise program, one should be evaluated by a physician to rule out any potential health risks. Pupils with physical restrictions or certain medical conditions may require an exercise program supervised by a healthcare professional, such as a physical therapist or exercise physiologist.
If dizziness, nausea, excessive shortness of breath or chest pain occur during any exercise program, the activity should be stopped, and a physician should be consulted before the affected resumes the activity. Persons who use any type of exercise equipment should be supervised by a knowledgeable fitness professional, such as a personal trainer.

Preparation for exercise
A physical examination by a physician is important to determine if strenuous exercise is appropriate or detrimental. Before beginning exercise, a proper warm-up is necessary to help prevent the possibility of injury resulting from tight muscles, tendons, ligaments and joints. Appropriate warm-up exercises include walking, light callisthenics, and stretching.

Aftercare
Proper cool-down after exercise is important and should include a gradual decrease in exercise intensity to slowly bring the heart rate back to the normal range, followed by stretches to increase flexibility and reduce the likelihood of muscle soreness. Following vigorous activities that involve sweating, lost fluids should be replaced by drinking water.

Risks
Improper warm-up and inappropriate use of weights can lead to muscle strains. Overexertion without enough time between exercise sessions to recuperate also can lead to muscle strains, resulting in inactivity due to pain. Perform high-impact activities, such as running may lead to stress, fractures of bones and may induction of asthma may occur. Dehydration is a risk during longer activities that involve sweating; it's advisable to always supply the body with water during and after the activity.

Normal results
Significant health benefits are obtained by including at least a moderate amount of physical exercise for 30 to 60 minutes daily. Regular physical activity plays a positive role in preventing disease and improving overall health status in a university environment. For teenagers and adults just beginning an exercise program, results (including weight loss, increased muscle strength, and aerobic capacity) will be noticeable in four to six weeks.

Personal/individual concerns
Given the increasing prevalence of overweight and obesity in teenagers and adults one needs to perform regular exercise and also serve as role models to other persons within their age gap. Television, computers, and video games have replaced physical activity for playtime for the majority of university students.

As a university student, one should commit to replacing sedentary activities with active indoor and outdoor games. For busy individuals during busy times e.g., exam times, exercise can be performed in multiple 10- to 15-minute sessions throughout the day.

How regular exercise benefits your studies at the University

A Healthy Body Equals a Healthy Mind: While you're at university, finding the time to exercise will not only improve your physical health but will also have a huge impact on other areas of your life. Whether you'll soon be starting university or you're already an established student, it's unlikely that you're thinking too much about staying in shape – unless you're studying sports science, in which case you're probably super-healthy already.

Studies show that students are known to engage in health-risking lifestyle behaviours, especially when it comes to food. We know that a kebab can seem appealing after a night out, but it's probably not the best idea if you're hitting the town five times a week. It can be difficult to find time to exercise when you're already trying to balance work, university and social life, but with plenty of accommodation options providing on-site gym facilities, it has never been easier for you to fit in a few workouts in and amongst everything else. Not only is exercising a great way to stay in shape while you study, but it can also make you an even better student than you already are. Some of the importance of exercise as a student in a university environment are;

1. Exercise improves your physical health

This is an obvious one, but exercise is a proven method when it comes to improving your physical health. Experts have found that there is a connection between being physically healthy and delivering a strong academic performance. This is because low-intensity exercise can give our energy levels a much-needed boost, which is perfect for when you're reaching the end of those back-to-back lectures and the fatigue kicks in.

2. Exercise promotes brain development

Studies have found that when you exercise, your body produces a protein called FNDC5, which is then released into the bloodstream. This then helps your brain to produce yet another protein called brain-derived neurotrophic factor (BDNF), which then prompts your body to grow new nerves and helps existing brain cells to survive – in other words, regular exercise makes your brain stronger.

Certain areas of the brain are more likely to develop in this way than others, but it just so happens that the hippocampus – the area of our brain that is involved with retaining information – is incredibly responsive to these proteins. This means that exercising regularly can help you take in and retain what you learn in lectures more easily than if you didn't exercise at all.

3. Exercise Improves Concentration

Research shows that just 20 minutes of exercise before studying can improve concentration and help you focus your learning. This is because intense physical activity causes blood to flow to the brain, which then fires up your neurones and promotes cell growth, particularly in the hippocampus (which is critical for learning).

Different types of exercise affect the brain in different ways, which means certain forms of physical activity can help you concentrate better than others. Start by taking a brisk walk to your next lecture, because this type of exercise has been linked to strong engagement when it comes to learning.

If you've got time before university, try out some yoga positions, as these are a great way to learn self-discipline and how to focus your mind. Most local gyms offer student discount on yoga classes, which means you can try it out for minimal cost!

4. Exercise Boosts Your Mood

It's important to remain enthusiastic throughout your studies, even when the work piles up. Trying to keep a positive mindset will enable you to stay focused and engaged through your lectures and exams, but it isn't always the easiest thing to do. Exercise has been found to treat mild-to-moderate depression as effectively as medication (without the side effects), so taking the time to exercise before you study will keep you in the right frame of mind for learning.

There can be times throughout university when the workload becomes a little bit stressful, which can impact your mood and cause you to lose focus. When stress affects the brain and its nerve connections, the rest of your body will feel the strain. Fortunately, this process can also work in reverse, so improving your physical health through exercise can have a positive impact on the brain and help to relieve stress.

5. Sport being part of exercise and fitness helps us to acknowledge and appreciate shared goal and common struggles

While many universities have excelled at welcoming fans in as sports spectators, colleges embrace the community service opportunity by inviting residents to participate as part of exercise and wellness programs. When we play, compete and sweat together (or volunteer our time and labour to create opportunities for others), strangers become neighbours and communities thrive. A university campus serves as a community centre, not an ivory tower; a welcoming gathering place and dynamic hub of engagement and activity.

6. Students who exercise tend to maintain a routine in their daily activities.

They use the time spent exercising as a break in their day and as a means to gather their thoughts. Students who participate in club sports or intramural activities can socialize and make friends have during their college years and beyond.

7. Sport as part of the exercise can help students to create a family away from home

For many if not most, athletics is a huge part of their identity. Our society places an immense value on athletic talent, which serves to affirm an athlete’s identity as they grow up. According to the NCAA, among the students who do exercise and identify their sporting talent, about six per cent of them make it to any collegiate level of competition and among collegiate athletes about one per cent have a career in major professional sports.

NON-COMMUNICABLE DISEASES

A non-communicable disease (NCD) or chronic diseases are diseases that are non-infectious (not transmissible directly from one person to another). They are generally of long duration and slowly progressive. NCDs include Parkison’s disease, autoimmune diseases, strokes, most cardiovascular/heart diseases, most cancers, non-insulin dependent diabetes mellitus, chronic kidney disease, osteoarthritis, osteoporosis, Alzheimer’s disease, cataracts, obesity, hypertension

Risk factors

Risk factors such as a person's background; lifestyle and environment are known to increase the likelihood of certain non-communicable diseases. They include age, gender, genetics, environmental factor (e.g., exposure to air pollution) and behaviors (e.g., smoking, unhealthy diet and physical inactivity). Most NCDs are considered preventable because they are caused by modifiable risk factors.

The WHO's identified five important risk factors for non-communicable disease in the top ten leading risks to health. These are raised blood pressure, raised cholesterol, tobacco use, alcohol consumption, and being overweight. The other factors associated with higher risk of NCDs include a person's economic and social conditions. It has been estimated that if the primary risk factors were eliminated, 80% of the cases of heart disease, stroke and type 2 diabetes and 40% of cancers could be prevented. Efforts focused on better diet and increased physical activity have been shown to control the prevalence of NCDs .

Influence of environment on NCDs

Poor health contributes to chronic NCDs including mental health conditions among university students thus increasing the rate of students absenteeism in classes and presenteeism, or productivity lost from students coming to college and performing below normal standards.

Warmer temperatures associated with climate change increase the formation of tropospheric ozone, a main constituent of smog and contributor to cardiorespiratory disease, and are associated with longer pollen seasons and increased pollen production, intensifying allergic respiratory diseases such as asthma. Particulate air pollution is driving increases in cardiovascular diseases and associated mortality. We are also currently experiencing a global epidemic of over-nutrition characterized by excessive intake of the wrong foods largely driven by inadequate access to fruits, vegetables, fish, and nuts and seeds resulting in unprecedented rates of obesity, diabetes, and heart disease.

NCDs associated with health and environmental factors

NCDs include many environmental diseases covering a broad category of avoidable and unavoidable human health conditions caused by external factors, such as sunlight, nutrition, pollution, and lifestyle choices. These include:

1. **Cardiovascular disease (CVD)**

Cardiovascular diseases (CVDs) are a group of disorders of the heart and blood vessels. They include; coronary heart disease, cerebrovascular disease, peripheral arterial disease, rheumatic heart disease, congenital heart disease, deep vein thrombosis and pulmonary embolism.

Heart attacks and strokes are usually acute events and are mainly caused by a blockage that prevents blood from flowing to the heart or brain. The most common reason for this is a build-up of fatty deposits on the inner walls of the blood vessels that supply the heart or brain. Strokes can be caused by bleeding from a blood vessel in the brain or from blood clots.

Causes and risk factors for cardiovascular disease

The most important behavioural risk factors of heart disease and stroke are unhealthy diet, physical inactivity, tobacco use and harmful use of alcohol. The effects of behavioural risk factors may show up in individuals as raised blood pressure, raised blood glucose, raised blood lipids, and overweight and obesity. These "intermediate risks factors" can be measured in primary care facilities and indicate an increased risk of heart attack, stroke, heart failure and other complications.

Prevention

Prevention of cardiovascular disease can be achieved by practising regular exercise, by keeping to a balanced healthy diet, reduction of salt in the diet, eating more fruit and vegetables. Avoiding tobacco smoking, avoiding harmful use of alcohol, maintenance of an optimal blood pressure and normal LDL-cholesterol and glucose levels.

In addition, drug treatment of hypertension, diabetes and high blood lipids are necessary to reduce cardiovascular risk and prevent heart attacks and strokes among people with these conditions.

2. **Chronic obstructive pulmonary disease (COPD) caused by smoking tobacco**

COPD is a group of progressive lung diseases. The most common of these diseases are emphysema and chronic bronchitis. Many people with COPD have both of these conditions.

Causes and risk factors for COPD

Most people with COPD are at least 40 years old and have at least some history of smoking. The longer and more tobacco products you smoke, the greater your risk of COPD is. Your risk of COPD is even greater if you have asthma and smoke.

You can also develop COPD if you're exposed to chemicals and fumes in the workplace. Long-term exposure to air pollution and inhaling dust can also cause COPD.

Prevention

The best way to prevent COPD is to never start smoking, and if you smoke, quit. Talk with your doctor about programs and products that can help you quit. Also, stay away from secondhand smoke, which is smoke from burning tobacco products, such as cigarettes, cigars, or pipes. Secondhand smoke also is smoke that has been exhaled, or breathed out, by a person smoking.

For people with COPD who have trouble eating because of shortness of breath or being tired should follow a special meal plan, rest before eating, take vitamins and nutritional supplements.

3. **Diabetes mellitus type 2**

Type 2 diabetes is a chronic disease that affects millions of people worldwide. Uncontrolled cases can cause blindness, kidney failure, heart disease and other serious conditions.

Before diabetes is diagnosed, there is a period where blood sugar levels are high but not high enough to be diagnosed as diabetes. This is known as prediabetes.

It’s estimated that up to 70% of people with prediabetes go on to develop type 2 diabetes. Fortunately, progressing from prediabetes to diabetes isn’t inevitable.

Although there are certain factors you can’t change such as your genes, age or past behaviors there are many actions you can take to reduce the risk of diabetes.

Prevention

Eating foods high in refined carbs and sugar increases blood sugar and insulin levels, which may lead to diabetes over time, avoiding these foods may help reduce your risk. Performing physical activity on a regular basis can increase insulin secretion and sensitivity, which may help prevent the progression from prediabetes to diabetes. Drinking water instead of other beverages may help control blood sugar and insulin levels, thereby reducing the risk of diabetes.

Carrying excess weight, particularly in the abdominal area, increases the likelihood of developing diabetes, losing weight may significantly reduce the risk of diabetes. Smoking is strongly linked to the risk of diabetes, especially in heavy smokers, quitting has been shown to reduce this risk over time.

Following a ketogenic or very-low-carb diet can help keep blood sugar and insulin levels under control, which may protect against diabetes. Avoiding large portion sizes can help reduce insulin and blood sugar levels and decrease the risk of diabetes. Avoiding sedentary behaviors like excessive sitting has been shown to reduce your risk of getting diabetes.

Consuming a good fiber source at each meal can help prevent spikes in blood sugar and insulin levels, which may help reduce your risk of developing diabetes. Consuming foods high in vitamin D or taking supplements can help optimize vitamin D blood levels, which can reduce your risk of diabetes.

Minimizing processed foods and focusing on whole foods with protective effects on health may help decrease the risk of diabetes. Drinking coffee or tea may help reduce blood sugar levels, increase insulin sensitivity and reduce the risk of diabetes. The herbs curcumin and berberine increase insulin sensitivity, reduce blood sugar levels and may help prevent diabetes.

4. **Lower back pain caused by too little exercise**

Back pain is a very common problem and will affect many of us at some point during our lives. The good news is that in most cases it isn’t a serious problem, and it might just be caused by a simple strain to a muscle or ligament. As far as possible, it’s best to continue with your normal everyday activities as soon as you can and to keep moving. Being active and exercising won’t make your back pain worse, even if you have a bit of pain and discomfort at first. Staying active will help you get better.

Causes

Often back pain doesn’t have one simple cause but may be due to poor posture, lack of exercise resulting in stiffening of the spine and weak muscles, muscle strains or sprains.

Managing back pain

The most important things to do to treat back pain is to keep moving, continue with everyday activities and have a healthy lifestyle. Some people worry that if they have back pain, doing certain activities such as lifting things, twisting and turning might make their back pain worse. It’s important to remember that our backs and our spines are very strong and are designed to move, in fact, too much rest can make back pain worse.

Being active and continuing with your everyday activities as soon as possible, and as much as possible, will speed up your recovery. There’s also evidence to suggest that how you respond emotionally to having back pain has an important impact on how quickly you get better. The more positive you are, the more active you are, the quicker your back will get better. Remember, if you're ever struggling don't suffer in silence, talk to a healthcare professional.

5. **Malnutrition**

Malnutrition refers to getting too little or too much of certain nutrients. It can lead to serious health issues, including stunted growth, eye problems, diabetes and heart disease. Malnutrition includes undernutrition and overnutrition, both of which can lead to health problems and development of chronic disease conditions. Long-term effects of undernutrition include a higher risk of obesity, heart disease and diabetes.

Common causes of malnutrition

Malnutrition is a worldwide problem that can result from environmental, economic and medical conditions.

Common causes of malnutrition include: Food insecurity or a lack of access to sufficient and affordable food. Digestive problems and issues with nutrient absorption. Excessive alcohol consumption; Heavy alcohol use can lead to inadequate intake of protein, calories and micronutrients. Mental health disorders; Depression and other mental health conditions can increase malnutrition risk. Inability to obtain and prepare foods; Studies have identified being frail, having poor mobility and lacking muscle strength as risk factors for malnutrition.

Prevention

Preventing and treating malnutrition involves addressing the underlying causes. Research suggests that some of the most effective ways to prevent malnutrition include providing iron, zinc and iodine pills, food supplements and nutrition education to populations at risk of undernutrition.

In addition, interventions that encourage healthy food choices and physical activity for students at risk of overnutrition may help prevent overweight and obesity. You can also help prevent malnutrition by eating a diet with a variety of foods that include enough carbs, proteins, fats, vitamins, minerals and water.

Treating malnutrition, on the other hand, often involves more individualized approaches. If you suspect that you or someone you know is undernourished, talk to a doctor as soon as possible. A healthcare provider can assess the signs and symptoms of undernutrition and recommend interventions, such as working with a dietitian to develop a feeding schedule that may include supplements.

6. **Skin cancer caused by radiation from the sun**

Skin cancer is the most common form of cancer in the africa, and the number of cases continues to rise. It is the uncontrolled growth of abnormal skin cells. While healthy cells grow and divide in an orderly way, cancer cells grow and divide in a rapid, haphazard manner. This rapid growth results in tumors that are either benign (noncancerous) or malignant (cancerous). The main types of skin cancer include Basal cell carcinoma, squamous cell carcinoma and melanoma.

Causes skin cancer

Exposure to sun causes most of the wrinkles and age spots on our faces. Ultraviolet (UV) radiation from the sun is the number one cause of skin cancer, but UV light from tanning beds is just as harmful. Exposure to sunlight during the winter months puts you at the same risk as exposure during the summertime.

Cumulative sun exposure causes mainly basal cell and squamous cell skin cancer, while episodes of severe blistering sunburns, usually before age 30, can cause melanoma later in life. Other less common causes are repeated X-ray exposure, scars from burns or disease, and occupational exposure to certain chemicals. Ultraviolet A (UVA) and Ultraviolet B (UVB) rays also affect the eyes and the skin around the eyes. Sun exposure may lead to cataracts and cancer of the eyelids.

Prevention

Nothing can completely undo sun damage, although the skin can sometimes repair itself. So, it's never too late to begin protecting yourself from the sun. Your skin does change with age; for example, you sweat less and your skin can take longer to heal, but you can delay these changes by limiting sun exposure.

Maintaining healthy skin

- **a.** Stop smoking: People who smoke tend to have more wrinkles than nonsmokers of the same age, complexion, and history of sun exposure.
- **b.** Apply sunscreen with a sun protection factor (SPF) of 30 or greater 30 minutes before sun exposure and then every 2 to 3 hours thereafter. Reapply sooner if you get wet or perspire significantly.
- **c.** Select cosmetic products and contact lenses that offer UV protection.
- **d.** Wear sunglasses with total UV protection.
- **e.** Avoid direct sun exposure as much as possible during peak UV radiation hours between 10 am and 4 pm.
- **f.** Perform skin self-exams regularly to become familiar with existing growths and to notice any changes or new growths.
- **g.** Relieve dry skin using a humidifier at home, bathing with soap less often (instead, use a moisturizing body wash), and using a moisturizing lotion.

7. Obesity

Body mass index (BMI) is a calculation that takes a person’s weight and height into account to measure body size. In adults, obesity is defined as having a BMI of 30.0 or more according to the Centers for Disease Control and Prevention (CDC). Obesity is associated with a higher risk for serious diseases, such as type 2 diabetes, heart disease, and cancer.

Causes and risks of obesity

Eating more calories than you burn in daily activity and exercise, on a long-term basis can lead to obesity. Over time, these extra calories add up and cause weight gain. But it’s not always just about calories in and calories out, or having a sedentary lifestyle.

Genetics, which can affect how your body processes food into energy and how fat is stored.

Growing older, which can lead to less muscle mass and a slower metabolic rate, making it easier to gain weight

Not sleeping enough, which can lead to hormonal changes that make you feel hungrier and crave certain high-calorie foods

Pregnancy, as weight gained during pregnancy may be difficult to lose and might eventually lead to obesity

Certain health conditions can also lead to weight gain, which may lead to obesity. These include: Polycystic ovary syndrome (PCOS) i.e. a condition that causes an imbalance of female reproductive hormones. Prader-Willi syndrome, a rare condition present at birth that causes excessive hunger. Cushing syndrome, a condition caused by having high cortisol levels (the stress hormone) in your system. Hypothyroidism (underactive thyroid), a condition in which the thyroid gland doesn’t produce enough of certain important hormones. Osteoathritis (OA) and other conditions that cause pain that may lead to reduced activity.

Obesity can lead to more than simple weight gain. Having a high ratio of body fat to muscle puts strain on your bones as well as your internal organs. It also increases inflammation in the body, which is thought to be a risk factor for cancer. Obesity is also a major risk factor for type 2 diabetes, heart disease, high blood pressure, certain cancers (breast, colon and endometrial), stroke, gallbladder disease, fatty liver disease, high cholesterol, sleep apnea and other breathing problems, anthritis and infertility.

Who is at risk for obesity

A complex mix of factors can increase a person's risk for obesity.

Genetics: Some people have genes that make it difficult for them to lose weight.

Environment and community: Your environment at home, at school, and in your community can all influence how and what you eat, and how active you are.

You may be at a higher risk for obesity if you live in a neighborhood with limited healthy food options or with high-calorie food options, like fast-food restaurants. Haven't yet lea9rned to cook healthy meals. Don't think you can afford healthier foods. Haven't found a good place to play, walk, or exercise in your neighborhood

Psychological and other factors: Depression can sometimes lead to weight gain, as some people may turn to food for emotional comfort. Certain antidepressants can also increase the risk of weight gain. Quitting smoking is always a good thing, but quitting may lead to weight gain too. In some people, it may lead to excessive weight gain. For that reason, it's important to focus on diet and exercise while you're quitting, at least after the initial withdrawal period. Medications, such as steroids or birth control pills, can also raise your risk for weight gain.

Prevention

On a personal level, you can help prevent weight gain and obesity by making healthier lifestyle choices. Aim for moderate exercise like walking, swimming or biking for 20 to 30 minutes every day.

Eat well by choosing nutritious foods, like fruits, vegetables, whole grains and lean protein. Eat high-fat, high-calorie foods in moderation.

The Bottom Line

You have control over many of the factors that influence non-communicable diseases. Eating the right foods and adopting other lifestyle behaviors that promote healthy body levels will give you the best chance at avoiding and preventing non-communicable diseases.

GREEN TECHNOLOGY

Green Technology is the development and application of products, equipment, and systems used to conserve the natural environment and resources, which minimize and reduces the negative impact of human activities.

Why green technology

Green technology is not a new technology; instead, it has picked momentum with the introduction of sustainable development initiatives because it is an environmentally friendly technology and hence known as environmental technology or clean technology. It uses innovative methods to create environmentally friendly products. Moreover, Green technology is known as sustainable technology as it takes into account the long- and short-term impact something has on the environment.

The need for green technology arises because natural resources are declining and pollution has increased due to the abundant use of non-renewable sources. Green products and creativities are by definition, environmentally friendly, energy efficiency, recycling, health and safety concerns, renewable resources, and more all go into the making of a green product or technology.

Products, systems or equipment based on green technology satisfy the following features or characteristics.

- It should reduce or minimize the degradation of the natural environment around us. It should have zero or minimum emission of greenhouse gases.
- It should be safe to use and should promote a healthy environment for all the forms of life including humans, birds, animals, etc.
- It should help in the conservation of energy as well as natural resources such as solar, water, wind, etc.
- It should promote the use of renewable resources.

Major applications of green technology

i. **Energy sector**

South Africa is the African continent's leader when it comes to renewable energy capacity. It is home to several solar power plants, including the 96 MW Jasper plant, a photovoltaic facility with 325,360 panels that can supply power to 80,000 homes. When it comes to wind power, onshore installations in South Africa amounted to 2,085 MW in 2018, according to the Global Wind Energy Council. The South African Wind Energy Association states that wind power accounts for 52% of the country's renewable energy power supply, while the average size of a South African wind power facility is 93.5 MW.

ii. **Building sector**

The applications of green technology solutions within the construction industry are fast improving. For instance, buildings that incorporate solar energy systems use up to 70% less electricity than conventional systems, according to the U.S Department of Energy. Also, green technology helps reduce a business' carbon footprint, reduces waste, conserves water and consumes reduced energy as compared to traditional technologies.

iii. **Transport sector**

Here, it is commonly referred to as Green Transportation or Sustainable Transportation which comprises those modes of transportation that do not depend on diminishing natural resources like fossil fuels. These transportation modes rely on renewable energy sources. They also have a very low impact on the environment as these modes produce minimal or no greenhouse gas emission.

Strengths / advantages of green technology

The world has a fixed amount of natural resources, some of which have already been depleted or ruined. For example, household batteries and electronics often contain dangerous chemicals that pollute our environments. Green technology offers us the best hope to counteract the effects of climate change and pollution. It would be productive to list all the benefits of green technologies in a point wise manner with examples.

i. **It does not emit or emit less harmful substances into the environment**. For instance; the solar cell, which directly converts energy from natural light into electrical energy via the process of photovoltaic. Generating electricity from solar energy equates to less consumption of fossil fuels, as well as the reduction of pollution and greenhouse gas emissions.

ii. **Rejuvenating Ecosystems**. Green or Clean technology is also being used to breathe life into ecosystems that have sustained a lot of damage due to human involvement. Through the use of this technology, trees are replanted, waste is managed and recycled. This ensures that the affected ecosystem can start again, and this time remains conserved. This helps to ensure that a lot of plant and animal species don't go extinct.

iii. **It has become popular as consumers of technology** are becoming more environmentally conscious. This will give benefits to investors in the long run in certain areas. A country Like Uganda has favorable solar irradiation of 1,825 kWh/m² to 2,500 kWh/m² per year. Small solar applications are often used in rural electrification projects such as Solar Home Systems or solar water heating. Over 30,000 solar PV systems have already been installed in rural areas. This brings more applicability of renewable energy technologies as well as bringing profits to the investors. This had major impacts on rural electrification as well as investment benefits to the private sector.

iv. **It requires less cost for maintenance**. This reduces operating costs and hence overall cost in the long run. For instance, Solar panels generally require very little maintenance. They are very durable and should last around 25-30 years with no maintenance. The only maintenance one needs to perform is to wash them clean of dirt and dust two to four times a year. This basic cleaning routine ensures that the sun can shine brightly on the panel, maximizing the amount of light available to turn into electrical power. It is the contrast to oil refineries or other non-renewable energy sources that require too much maintenance and constantly, that if not done can lead to major destructive accidents.

v. As it uses renewable natural resources and hence we will never run out of vital resources such as water and electricity. With renewable energy sources, such as wind power and solar power such problems will be less likely to encounter the problem of fuel getting vanished.

Energy is being conserved through the use of such technology. Alternatives to devices that use a lot of electricity or fuel are being introduced to the public. The use of electric cars is on the rise, especially in the UK. People are using environment-friendly devices and appliances. While installation of such devices, namely solar panels, might be expensive for some people the benefits it offers with regards to reducing bill expenses are tremendous Green technology purifies the air, hence it will slow down the effects of global warming due to a reduction in CO_2 emissions. Generating electricity from solar and wind energy means less consumption of fossil fuels and a reduction in pollution and greenhouse gas emissions. Reducing and recycling of plastic waste is advantageous for the environment. Hence, trendy reusable water bottles that can be refilled are health-promoting, eco-friendly, and green.

vi. **Creates new avenues and Employment**. Green technologies have the potential to give birth to sectors that were previously not thought of. In this 21st century, we need to create paths that can improve the economy and the environment at the same time. Green Technology allows us to combine the two. We can see a very relevant example of that of waste disposal. Earlier waste management was only limited to waste dumping; today waste management. Is a $25 billion industry in South Asia alone? The green energy sector is responsible for a host of job opportunities on the market today. As a result, many employment options come up for people, and some of them include environmental managers, solar energy experts, and efficient lighting experts as well.

vii. **Recycling**. Green technology helps manage and recycle waste material. It allows it to be used for beneficial purposes. This technology is used for waste management, waste incineration, and more. A lot of recyclable material has allowed individuals to create plant fertilizer, sculptures, fuel, and even furniture.

viii. **Purifying of Water**. Green technology purifies water. The scarcity of pure drinking water is a major concern. Through the use of various technologies, a lot of campaigns have been successful in providing people with clean drinking water.

ix. **Green Farming**: Green approaches to farming have been proven to be not only healthier for humans but also productive for the soil. It leads to higher productivity over sustained periods contrary to the inorganic farming practices which lead to a decrease in yield after a certain period. Inorganic farming methods have had a very bad environmental impact and have resulted in degradation of the aquatic life of surface water bodies; it has also stripped the earth of the various insects and worms which helped in crop production. Fortunately, the effects of inorganic farming are yet to be seen in a big way but they are inevitable. Places, where organic (Green) farming is practiced, are already showing that it is a better approach in the long run.

x. **Green Buildings**: Green construction technologies are also coming up and are being encouraged in a big way. They have high initial investment but have a minimal environmental impact and are energy efficient. Because office and residential buildings consume a large share of the energy pie of any country they are certain to have significant benefits in the future over conventional buildings. Since they reduce energy consumption and wastage; these buildings can recover their cost over an acceptable time frame. Such constructions prove economical and eco-friendly in the long run and thus are beneficial to the individual and the society as a whole

xi. **A benefit to the urban areas:** Taking into account the current chaotic situation of the cities of the world one can easily argue that they need to take urgent environment improvement measures. Cities that actively pursued their environmental concerns in the last ten years are showing a marked improvement in their environment quality parameters. For example, Delhi launched CNG fuelled public transport in a phased manner and in December 2002, the last diesel bus was flagged off. This was done as a measure to improve the air quality of Delhi where the toxic gas levels were off the charts, sometimes exceeding 5-12 times the normal values. Since then Delhi has shown steady improvement in the air quality. The annual average level of restorable suspended particulate matter (RSPM or PM10) in residential areas was 143 micrograms per cubic meter. It dropped to 115 micrograms per cubic meter by 2005.

xii. **Benefits to the Rural Areas: G**reen technologies involve humans in a much bigger way than conventional technologies and thereby empower them by giving them responsibilities and avenues to gain, learn and progress. Green technologies have had a great impact on communities of the areas where they have been implemented. The provision of biogas plants to rural households has empowered communities and has increased their productivity. The same has been the case with the distribution of solar lanterns through certain programs. Similarly, Green technology requires more involvement. Hence it empowers people. It (green technologies) helps people as it can be diffused much more easily in remote areas due to its discretized nature.

WEAKNESSES / DISADVANTAGES OF GREEN TECHNOLOGY

Green technologies have been applied in many sectors however they have not been put into complete use e.g. in the energy sector they are still "alternate sources of energy". Since we have not seen them in usage full time we cannot observe what are their disadvantages. However certain concerns have been already raised, these are related to the reliability of these technologies, convenience of their usage, the investment required, their hypocritical use, ethical and social issues, etc. We will discuss these issues one by one.

i. **The initial investment or implementation cost is very high.** For instance; India's upper limit of the installation cost of a 1 MW photovoltaic solar power plant is roughly Rs 300-350 million (discounting the government grants). On the other hand, the installation cost of a subcritical coal power plant (Ultra mega power plant 4000MW) is roughly Rs 184,736 million. i.e. Rs 46.184 million per MW. The difference is extremely large and quite unaffordable for many developing countries; large investments in green technologies by the government would effectively slow down the cash flow in other important sectors e.g. health care and infrastructure.

ii. **Social downside**: Socially green technologies are still to become popular. This is more connected to the technical assistance and incentives of usage. Social downsides are extremely area specific. For instance in rural areas; people lack expertise, they also lack the proximity of servicing as well as sensitization and awareness creation opportunities. For urban areas, sometimes; the solar water heaters fail to work due to week insolation like in winter seasons hence people opt for electrical heaters which are always active as long as there is power supply constantly.

iii. **The technology is still evolving and many of the products are at the R&D stage**. Hence people are unaware of performance results. For instance; today solar panels are being manufactured at a prolific rate, however, it is not certain what will happen to the panels once they finish their life cycle. The disposal of solar panels is still in the R&D stage. Many solar panels contain toxic matter which can be dangerous; this problem will surface after two decades if we are not prepared by then.

iv. **Technical Downside:** To be popular any technology has to be reliable. Some green technologies have severe drawbacks in this area. For instance; solar power plants work fine on a sunny day but their performance becomes miserable the moment solar radiation on the surface drops, this cannot be afforded in today's competitive world. Similarly, solar air conditioning fails in the rainy season when there is a lot of humidity and low solar radiation. These drawbacks push a person to install conventional technologies for his/her usage. We can hope that these flaws will be removed once these technologies mature with time. Power savers meant to improve the efficiencies of tube lights etc. themselves fail very early giving the use of little incentive to use them. This causes significant operational discomfort to the user.

v. **Ethical Downside:** Today the world is in a very complicated state where ethics have started clashing, many times environmental ethics clash with general human ethics. A very relevant example is when USA started to use corn to make ethanol for bio-diesel. How ethical is it to use food to create fuel for cars when millions are hungry no matter how environment friendly the technology is? This argument effectively stalled this project. Green buildings, solar power plants all are currently very expensive; how justified is their implementation when it is more necessary to feed a hungry population? Can a poor person be forced to purchase new fuel efficient kitchen equipment when he barely makes his ends meet? There are umpteen ethical issues that spring up and are mostly related to cost versus survival questions. Prices of green technologies will go down in the future, but the basic problem is that their prices will go down once we make the initial investment but doing the initial investment is not easy. This is a vicious circle and is perhaps the biggest difficulty in the implementation of green technologies.

vi. **Land use:** A lot of green energy sources use more land than other non-green sources, which is a big negative if land use, space available, and arable land are important (depending on the location). For instance, A 12 KW home generator is only like 28 inches x 26 inches…a similar sized solar panel would only generate like 100 watts (0.1 KW) (Alice, What is your view on the conversion of land from food production into energy crop production?

vii. **Green energy is still focused on fixed base consumption**. By this I mean its common applications are residential and commercial power. Green technology in the transportation sector is less mature. Although this is changing rapidly, battery technology still has a ways to go down the technology evolution path. Fuel cells are a potential solution, but the energy tradeoffs to generate the hydrogen for the cells still need some efficiency improvements. If it comes in greater availability it will allow greater ranges for electric vehicles.

Along with any energy solution(fossil or green) that we use in the future, we will require a 'smart grid' or electrical grid that allows the more efficient transmission of power over longer distances, as well as, storage of excess power during periods of low demand. With a smart grid, any limitations of generation points vs. consumption points disappear. This is still lacking hence hindering green technology's applicability.

We might end up seeing many effects that we cannot think as of now because we still have not experienced green technologies in a big way. We cannot say for certain that green technologies will be good for us in all ways in the distant future, but one thing is for certain that we will be risking the fate of mankind if we do not adopt them in the present. Today green technologies are the need of the hour without any doubt as they are most appropriate to our current world needs.

GREENING IN AGRICULTURE AND ENVIRONMENT

Greening it activities such as protecting permanent grassland designated as environmentally sensitive grassland, growing a minimum number of crops. Farming five per cent of your arable area in a manner that promotes biodiversity, i.e. Ecological Focus Area.

The aims of greening

The farmers act as managers of the countryside, they shape landscapes and through their work, farmers provide public goods beneficial to all. The "green direct payment" (or "greening") supports farmers who adopt or maintain farming practices that help meet environmental and climate goals.

Green economy

In a green economy, growth in employment and income are driven by public and private investment into such economic activities, infrastructure and assets that allow reduced carbon emissions and pollution, enhanced energy and resource efficiency, and prevention of the loss of biodiversity and ecosystem services. The benefits of green economy in businesses are; enhanced brand image and increased competitive advantage, increased productivity and reduced costs, better financial and investment opportunities, increased preparedness for future legislation and costs, improved recruitment and retention of quality employees, and a healthier work environment for employees.

RENEWABLE ENERGY

Renewable energy comes from sources or processes that are constantly replenished. These sources of energy include solar energy, wind energy, geothermal energy, and hydroelectric power.

Renewable sources are often associated with green energy and clean energy, but there are some subtle differences between these three energy types. Where renewable sources are those that are recyclable, clean energy are those that do not release pollutants like carbon dioxide, and green energy is that which comes from natural sources. While there is often cross-over between these energy types, not all types of renewable energy are actually fully clean or green. For example, some hydroelectric sources can actually damage natural habitats and cause deforestation.

TYPES OF RENEWABLE ENERGY

There are a range of renewable energy sources that have been developed, with each offering their own advantages and challenges depending on factors such as geographical location, requirements for use and even the time of year.

1. Solar Power

The potential for the sun to supply our power needs is huge, considering the fact that enough energy to meet the planet's power needs for an entire year reaches the earth from the sun in just one hour. However, the challenge has always remained in how to harness and use this vast potential.

We currently use solar energy to heat buildings, warm water and power our devices. The power is collected using solar, or photovoltaic (PV), cells made from silicon or other materials. These cells transform sunlight into electricity and can power anything from the smallest garden light to entire neighborhoods'. Rooftop panels can provide power to a home, while community projects and solar farms that use mirrors to concentrate the sunlight can create much larger supplies. Solar farms can also be created in bodies of water, called 'floatovoltaics' these provide another option for locating solar panels. As well as being renewable, solar powered energy systems are also clean energy sources, since they don't produce air pollutants or greenhouse gases. If the panels are responsibly sited and manufactured they can also count as green energy as they don't have an adverse environmental impact.

2. Wind Power

Wind energy works much like old-fashioned windmills did, by using the power of the wind to turn a blade. Where the motion of these blades would once cause millstones to grind together to make flour, today's turbines power a generator, which produces electricity.

When wind turbines are sited on land they need to be placed in areas with high winds, such as hilltops or open fields and plains. Offshore wind power has been developing for decades with wind farms providing a good solution for energy generation while avoiding many of the complaints around them being unsightly or noisy on land. Of course, offshore use has its own drawbacks due to the aggressive environments the turbines need to operate in.

3. Hydroelectric Power

Hydroelectric power works in a similar manner to wind power in that it is used to spin a generator's turbine blades to create electricity. Hydro power uses fast moving water in rivers or from waterfalls to spin the turbine blades and is widely used in some countries. It is currently the largest renewable energy source in the United States, although wind energy is fast closing the gap.

Hydroelectric dams are a renewable energy source, but these are not necessarily green energy sources. Many of the larger 'mega-dams' divert natural water sources, which creates a negative impact for animal and human populations due to restricted access to the water source. However, if carefully managed, smaller hydroelectric power plants (under 40 megawatts) do not have such a catastrophic effects on the local environment as the divert just a fraction of the water flow.

4. Biomass Energy

Biomass energy uses organic material from plants and animals, including crops, trees, and waste wood. This biomass is burned to create heat which powers a steam turbine and generates electricity. While biomass can be renewable if it is sustainably sourced, there are many instances where this is neither green nor clean energy.

Studies have shown that biomass from forests can produce higher carbon emissions than fossil fuels, while also have an adverse impact on biodiversity. Despite this, some forms of biomass do offer a low-carbon option given the correct circumstances. Sawdust and wood chippings from sawmills, for example, can be used for biomass energy where it would normally decompose and release higher levels of carbon into the atmosphere.

6. Geothermal

Geothermal energy uses the heat trapped in the Earth's core which is created by the slow decay of radioactive particles in rocks at the centre of the planet. By drilling wells, we are able to bring highly heated water to the surface which can be used as a hydrothermal resource to turn turbines and create electricity. This renewable resource can be made greener by pumping the steam and hot water back into the earth, thereby lowering emissions.

The availability of geothermal energy is closely tied to geographical location, with places such as Iceland having an easily reached, ready supply of geothermal resources.

7. Tidal Power

Tidal power offers a renewable power supply option, since the tide is ruled by the constant gravitational pull of the moon. The power that can be generated by the tide may not be constant, but it is reliable, making this relatively new resource an attractive option for many.

However, care needs to be taken with regard to the environmental impact of tidal power, as tidal barrages and other dam-like structures can harm wildlife.

Advantages of renewable energy

Renewable energy offers a range of benefits including offering a freely available source of energy generation. As the sector grows there has also been a surge in job creation to develop and install the renewable energy solutions of tomorrow. Renewable sources also offer greater energy access in developing nations and can reduce energy bills too.

Of course, one of the largest benefits of renewable energy is that much of it also counts as green and clean energy. This has created a growth in renewable energy, with wind and solar being particularly prevalent.

However, these green benefits are not the sole preserve of renewable energy sources. Nuclear power is also a zero-carbon energy source, since it generates or emits very low levels of CO_2. Some favour nuclear energy over resources such as solar and wind, since nuclear power is a stable source that is not reliant on weather conditions. Which brings us onto some of the disadvantages of renewable energy.

Disadvantages of renewable energy

As mentioned above, many renewable energy sources cannot be relied upon all the time. When the sun goes down or hides behind a cloud, we cannot generate solar power and when the wind doesn't blow, we cannot create enough wind energy. For this reason, fossil fuels are still in use to top-up renewable sources in many countries.

This variable production capacity means that large energy storage solutions are required to ensure there is enough power when renewable energy generation dips. An alternative solution is to deploy several renewable technologies, creating a more flexible system of supply that can counteract dips in production for a given source.

Some renewable resources, such as hydropower and biomass, do not suffer with these problems of supply, but these both have their own challenges related to environmental impact, as noted above. In addition to this, some renewable energy sources, such as solar and wind farms, create complaints from local people who do not want to live near them.

However, this is not always the case, as shown by the example of Ardossan Wind Farm in Scotland, where most local residents believe the farm enhanced the area. Furthermore, a study by the UK Government found that, "projects are generally more likely to succeed if they have broad public support and the consent of local communities. This means giving communities both a say and a stake." This theory has been proven in Germany and Denmark, where community-owned renewable projects have proven popular.

BUILT ENVIRONMENT

The term built environment refers to the human-made surroundings that provide the setting for human activity, ranging in scale from buildings and parks or green space to neighborhoods and cities that can often include their supporting infrastructure, such as water supply or energy networks.

The built environment is a material, spatial, and cultural product of human labor that combines physical elements and energy in forms for living, working, and playing. It has been defined as "the human-made space in which people live, work, and recreate on a day-to-day basis"

A built environment is developed in order to satisfy residents' requirements. Human needs can be physiological or social and are related to security, respect, and self-expression. People want their built environment to be aesthetically attractive and to be in an accessible place with a well-developed infrastructure, convenient communication access, and good roads, and the dwelling should also be comparatively cheap, comfortable, with low maintenance costs, and have sound and thermal insulation of walls.

People are also interested in ecologically clean and almost noiseless environments, with sufficient options for relaxation, shopping, fast access to work or other destinations, and good relationships with neighbors.

The study of the built environment is interdisciplinary in nature and can include such disciplines as:

- visual arts,
- architecture,
- engineering,
- urban planning,
- history,
- interior design,
- industrial design,
- geography,
- environmental studies,
- anthropology / sociology

It must be admitted that the most serious problems of built environments (eg, unemployment, vandalism, lack of education, robberies) are not always related to the direct physical structure of housing. Increasing investment into the development of social and recreational centers, such as athletic clubs, physical fitness centers, and family entertainment centers, the infrastructure, a good neighborhood and better education of young people, can solve such problems. Investment, purchase and sale of a property, and its registration have related legal issues. The legal system of a country aims to reflect its existing social, economic, political, and technical state and the requirements of the market economy. As illustrated, the life cycle of the built environment can be assessed taking into account many quantitative and qualitative criteria. Life cycle of the built environment quantitative and qualitative analyses aspects are presented in Fig. 1.

Fig. 1. Life cycle of the built environment quantitative and qualitative analyses aspects.

Each one of these Level 1 aspects subsystems (see Fig. 1) based on the principle of a tree diagram can be discussed in much greater detail. To illustrate, the life cycle of the built environment described according to an example of the i^{th} level aspects subsystem could be energy. In view of the global practice, any analysis of various aspects characteristic to the life cycle of the built environment focuses on the analysis of energy. Life cycle of the energy-efficient built environment application areas are presented in Fig. 2.

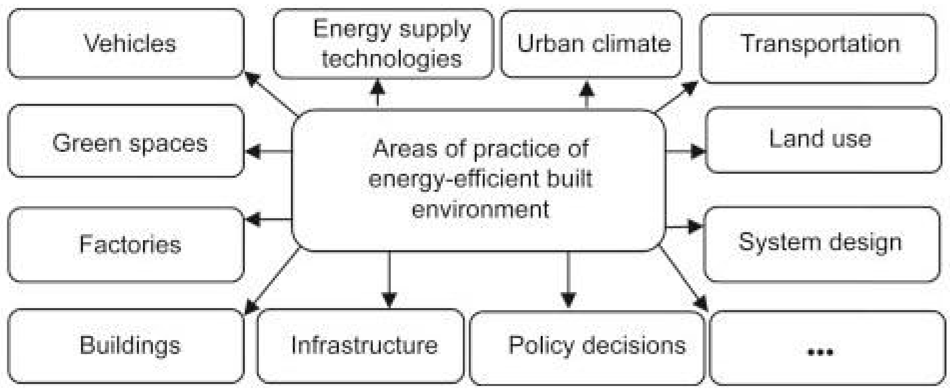


Fig. 2. Areas of practice of energy-efficient built environment.

Here is an overview of the applications used in the built environment:

- ***Domotic and home automation:*** Energy and water use (energy and water supply consumption monitoring to obtain advice on how to save cost and resources), remote control appliances (switching on and off appliances remotely to avoid accidents and save energy), intrusion detection systems (detection of window and door openings and violations to prevent intruders), art and goods preservation (monitoring of conditions inside museums and art warehouses).
- ***Smart cities:*** Smart parking (monitoring of parking spaces availability in the city), structural health (monitoring of vibrations and material conditions in buildings, bridges, and historical monuments), noise urban maps (sound monitoring in bar areas and centric zones in real-time), electromagnetic field levels (measurement of the energy radiated by cell stations and WiFi routers), traffic congestion (monitoring of vehicles and pedestrian levels to optimize driving and walking routes), smart lighting (intelligent and weather-adaptive lighting in street lights), waste management (detection of rubbish levels in containers to optimize the trash collection routes), smart roads (intelligent highways with warning messages and diversions according to climate conditions and unexpected events like accidents or traffic jams).
- ***Smart environment:*** Forest fire detection (monitoring of combustion gases and preemptive

fire conditions to define alert zones), air pollution (control of CO_2 emissions of factories, pollution emitted by cars), snow level monitoring (snow level measurement to know in real time the quality of ski tracks and allow security corps avalanche prevention), landslide and avalanche prevention (monitoring of soil moisture, vibrations, and earth density to detect dangerous patterns in land conditions), earthquake early detection (distributed control in specific places of tremors).

- ***Smart water:*** Potable water monitoring (monitor the quality of tap water in cities),
  chemical leakage detection in rivers (detect leakages and wastes of factories in rivers), swimming pool remote measurement (control remotely the swimming pool conditions), pollution levels in the sea (control real-time leakages and wastes in the sea), water leakages (detection of liquid presence outside tanks and pressure variations along pipes), river floods (monitoring of water level variations in rivers, dams, and reservoirs).
- ***Smart metering:*** Smart grid (energy consumption monitoring and management), tank level (monitoring of water, oil, and gas levels in storage tanks and
  cisterns), photovoltaic installations (monitoring and optimization of performance in solar energy plants), water flow (measurement of water pressure in water transportation systems), silos stock calculation (measurement of emptiness level and weight of the goods).
- ***Security and emergencies:*** Perimeter access control (access control to restricted areas and detection of people in non-authorized areas), liquid presence (liquid detection in data centers, warehouses, and sensitive building grounds to prevent breakdowns and corrosion),
  radiation levels (distributed measurement of radiation levels in nuclear power stations surroundings to generate leakage alerts), explosive and hazardous gases (detection of gas levels and leakages in industrial environments, surroundings of chemical factories, and inside mines).
- ***Retail:*** Supply-chain control (monitoring of storage conditions along the supply chain and product tracking for traceability purposes), NFC payment (payment processing based on
  location or activity duration for public transport, gyms, theme parks, etc.), intelligent shopping applications (getting advice in the point of sale according to customer habits, preferences, presence of allergic components for them, or expiring dates), smart product management (control of rotation of products in shelves and warehouses to automate restocking processes).